Module 3: (8 Lectures)

- Highway Materials: Soil relevant properties, Various tests,
- Aggregates – strength, hardness, toughness, soundness, durability, shape, specific gravity, water absorption
- Bituminous materials – Bitumen, Tar, and Asphalt – various properties
- Design of Bituminous paving mixes-Marshall stability test

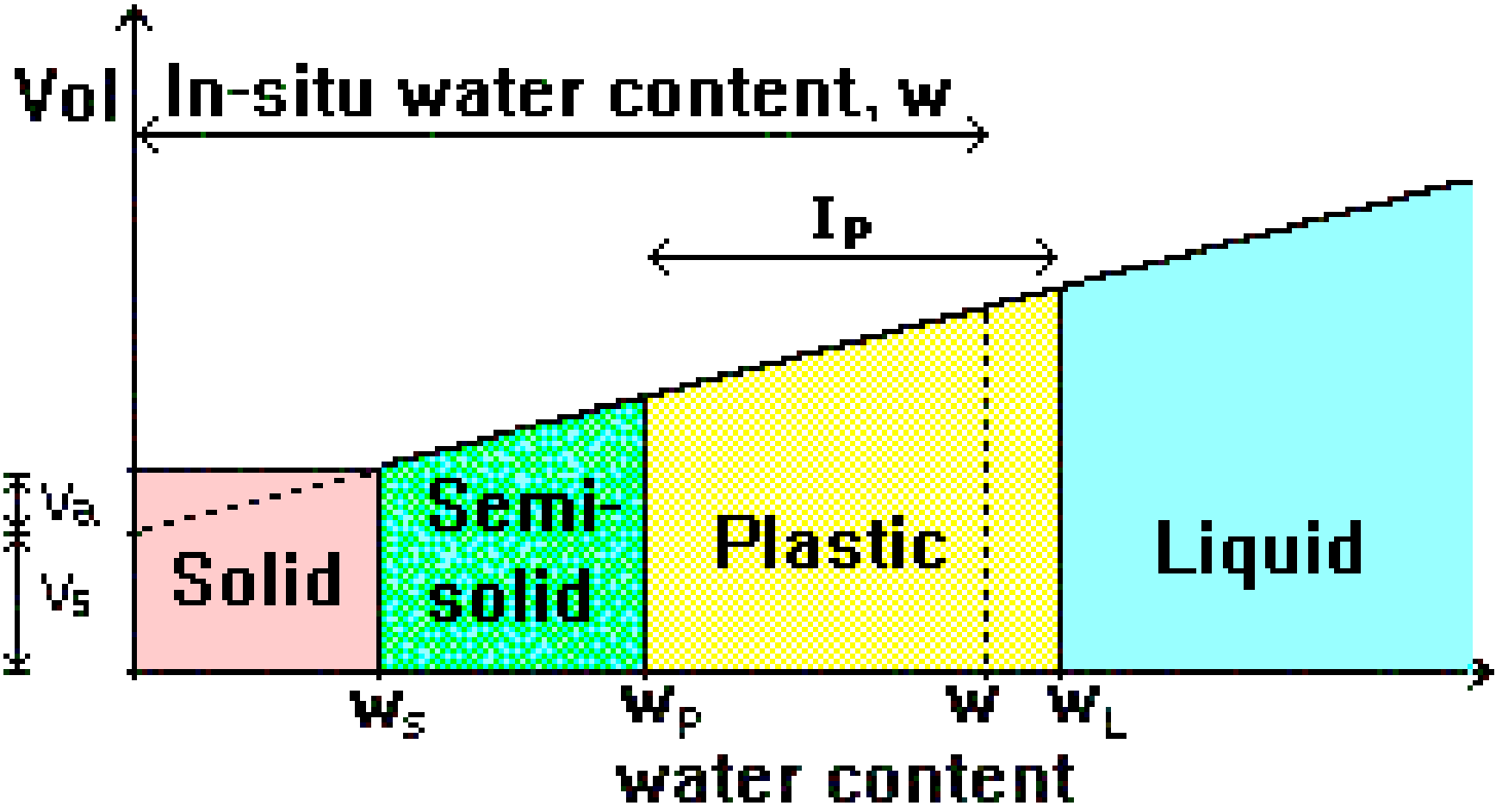
Highway Materials: Soil – Properties and Tests

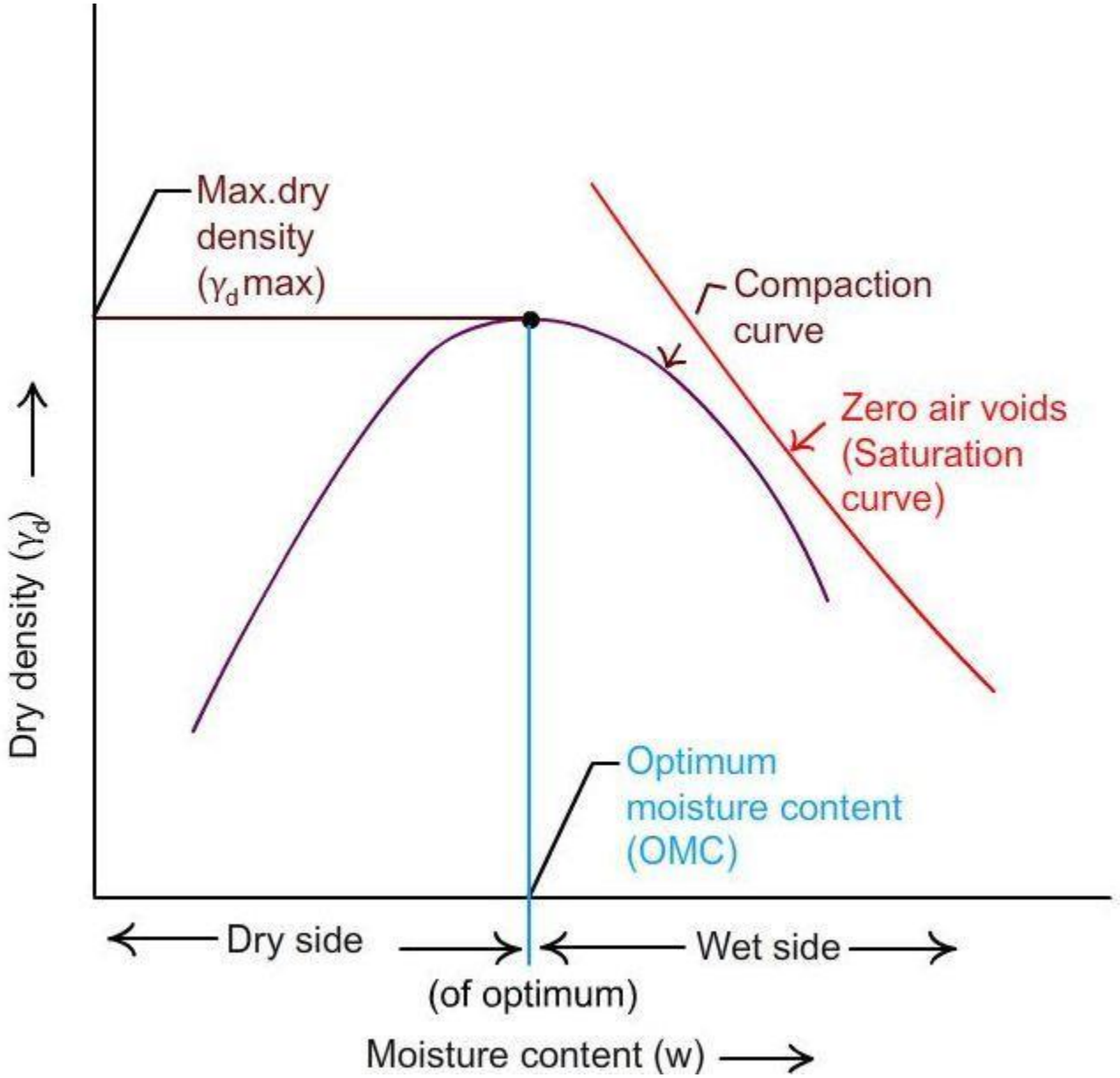
- Soil is a fundamental material in highway construction, **forming the subgrade and affecting the overall pavement performance.**

- Understanding **soil properties** and conducting **appropriate tests** are **essential for ensuring a strong and stable foundation**.

Soil Properties for Highway Engineering

Property	Definition	Use in Highway Engineering	Formula (if applicable)
Grain Size Distribution	Percentage of different particle sizes in soil.	Determines soil classification (coarse or fine-grained) and suitability for pavement layers.	Uniformity Coefficient (Cu) = D60/D10 Coefficient of Gradation (Cc) = $(D30^2) / (D10 \times D60)$
Moisture Content (w)	The amount of water present in soil, expressed as a percentage of dry weight.	Affects compaction and strength of soil.	$w (\%) = [(Ww - Wd) / Wd] \times 100$
Specific Gravity (G)	Ratio of the density of soil solids to the density of water.	Used in void ratio, density, and permeability calculations.	$G = \rho s / \rho w$ (ρs = Soil density, ρw = Water density)
Atterberg Limits	Defines plasticity and moisture sensitivity of fine-grained soils.	Helps classify soil type and determines workability.	–
Liquid Limit (LL)	Moisture content at which soil changes from plastic to liquid state.	Used to classify clayey soils and evaluate their compressibility.	–
Plastic Limit (PL)	Moisture content at which soil starts to behave as a plastic material.	Helps in determining soil plasticity and its engineering behavior.	–
Plasticity Index (PI)	Difference between Liquid Limit and Plastic Limit.	Indicates soil's ability to undergo deformation without cracking.	PI = LL - PL
Shrinkage Limit (SL)	Moisture content below which soil does not shrink further.	Important for preventing pavement cracks due to volume change.	–
Compaction Characteristics	Relationship between soil density and moisture content.	Helps achieve required soil strength and stability.	Dry Density (γd) = $\gamma b / (1 + w)$
Maximum Dry Density (MDD)	The highest dry density achieved at optimum moisture content.	Used for proper compaction of subgrade.	–


Vol
In-situ water content, w
Ip
Va
Vs
Solid
Semi-solid
Plastic
Liquid
ws
wp
w
wL
water content


Max.dry
density
($\gamma_d$max)
Compaction
curve
Zero air voids
(Saturation
curve)
Dry density (γ_d)
Optimum
moisture content
(OMC)
Dry side
Wet side
(of optimum)
Moisture content (w)

Property	Definition	Use in Highway Engineering	Formula (if applicable)
Optimum Moisture Content (OMC)	Moisture content at which maximum dry density is achieved.	Ensures effective compaction and strength of soil.	–
California Bearing Ratio (CBR)	Ratio of force per unit area required to penetrate a soil mass with a standard plunger.	Determines subgrade strength for pavement design.	CBR (%) = (Load at penetration / Standard Load) × 100
Shear Strength (τ)	Resistance of soil against shear failure.	Determines load-bearing capacity of soil.	$\tau = c + \sigma \tan(\phi)$ (c = Cohesion, σ = Normal stress, ϕ = Angle of internal friction)
Permeability (k)	Ability of soil to allow water flow.	Determines drainage capacity of subgrade.	$k = (QL) / (Aht)$ (Q = Discharge, L = Length, A = Area, h = Head difference, t = Time)
Void Ratio (e)	Ratio of void volume to volume of soil solids.	Important in compaction and settlement analysis.	$e = (Vv / Vs)$
Porosity (n%)	Percentage of void space in a soil sample.	Affects permeability and drainage.	$n (\%) = (Vv / Vt) \times 100$
Degree of Saturation (S)	Percentage of voids filled with water.	Helps in determining soil moisture behavior.	$S (\%) = (Vw / Vv) \times 100$
Unconfined Compressive Strength (UCS)	Maximum axial stress soil can withstand under no lateral confinement.	Determines strength of cohesive soils.	$UCS = P / A$ (P = Load, A = Cross-sectional area)

Classification of Soil in Highway Engineering

1) Textural Soil Classification

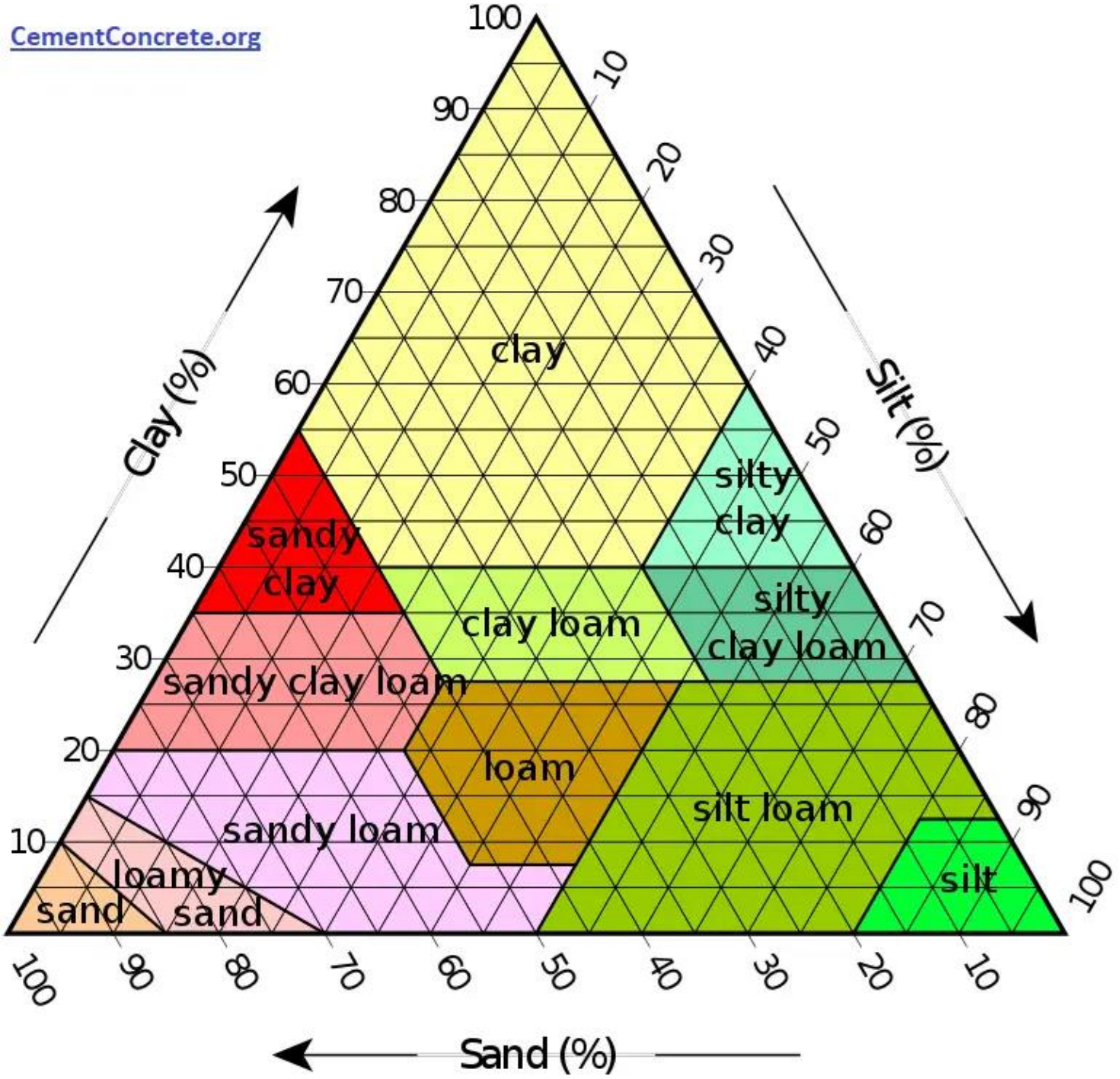
This classification is based on the **percentage of sand, silt, and clay** in the soil. It is commonly represented using the **textural triangle** developed by the **USDA (United States Department of Agriculture).**

📌 **Use in Highway Engineering:** Determines **drainage properties** and **suitability for subgrade** based on texture.

Categories of Soil (Based on Particle Size Distribution)

Soil Type	Sand (%)	Silt (%)	Clay (%)
Sand	>85	<15	<10
Sandy Loam	50-85	15-50	<20
Loam	25-50	30-50	10-30
Silt Loam	<20	50-80	10-30
Clay Loam	<45	20-50	30-40
Clay	<20	<40	>40

CementConcrete.org



2) Unified Soil Classification System (USCS)

Developed by **Casagrande and later modified by ASTM**, USCS is widely used in **geotechnical and highway engineering.**

📌 **Use in Highway Engineering:** Determines **suitability of soil for embankments, subgrade stability, and drainage properties.**

Major Soil Divisions in USCS:

Symbol	Soil Type	Description
Coarse-Grained Soils (More than 50% retained on 0.075 mm sieve)		
GW	Well-graded gravel	Good load-bearing capacity, excellent drainage
GP	Poorly graded gravel	Poor gradation, low cohesion
GM	Silty gravel	Moderate strength, drainage varies
GC	Clayey gravel	Higher cohesion, poor drainage
SW	Well-graded sand	Good stability, excellent drainage
SP	Poorly graded sand	Uniform grain size, poor stability
SM	Silty sand	Moderate stability
SC	Clayey sand	Cohesive, low permeability
Fine-Grained Soils (More than 50% passes through 0.075 mm sieve)		
ML	Low-plasticity silt	Low strength, compressible
CL	Low-plasticity clay	Good strength, low permeability
OL	Organic silt/clay	Poor strength, highly compressible
MH	High-plasticity silt	High compressibility, poor drainage
CH	High-plasticity clay	High strength, low permeability
OH	Organic clay	Very poor strength, high shrinkage
PT	Peat	High organic content, unsuitable for construction

3) BIS Soil Classification (Indian Standard – IS: 1498-1970)

The **Bureau of Indian Standards (BIS)** classification system is based on **particle size and plasticity characteristics.**

Major Soil Groups as per BIS Classification:

📌 **Use in Highway Engineering:** The **BIS system is widely used in India** for subgrade evaluation, embankment design, and pavement construction.

Soil Group	Sub-Classification	Characteristics
Coarse-Grained Soils (More than 50% retained on 75-micron sieve)	Gravels (G)	Well-graded (GW), Poorly graded (GP), Silty (GM), Clayey (GC)
	Sands (S)	Well-graded (SW), Poorly graded (SP), Silty (SM), Clayey (SC)
Fine-Grained Soils (More than 50% passes 75-micron sieve)	Silt (M)	Low-plasticity (ML), High-plasticity (MH)
	Clay (C)	Low-plasticity (CL), High-plasticity (CH)
Highly Organic & Other Soils	Peat (Pt)	Organic soils with high compressibility

4) Highway Research Board (HRB) Soil Classification (AASHTO Classification)

Developed by the **American Association of State Highway and Transportation Officials (AASHTO)**, this system categorizes soil based on **grain size distribution, plasticity index, and load-bearing capacity** (CBR).

HRB-CLASSIFICATION OF SOILS AND SOIL-AGGREGATE MIXTURES

General Description	*Granular materials (35% or less passing 75 micron IS sieve)*							*Silt clay materials (more than 35% passing 75 micron IS sieve)*			
Group Classification	A–1		A–3	A–2				A–4	A–5	A–6	A–7
	A–1–*a*	A–1–*b*		A–2–4	A–2–5	A–2–6	A–2–7				A–7–5 A–7–6
Sieve analysis, percent passing											
2.0 mim IS sieve	50 max										
425 micron sieve	30 max	50 max	51 min								
75 micron sieve	15 max	25 max	10 max	35 max	35 max	35 max	35 max	36 min	36 min	36 min	36 min
Characteristics of fraction passing 425 micron sieve											
Liquid Limit				40 max	41 min	40 max	41 min	40 max	41 min	40 max	41 min
Plasticity Index	6 max		NP	10 max	10 max	11 min	11 max	10 max	10 max	11 min	11 min
Group Index	Zero					4 max		8 max	12 max	16 max	20 max
Usual type of significant constituent materials	Stone fragments gravel and sand		Fine sand	Silty or clayey gravel and sand				Silty soils		Clayey soils	
General rating as subgrade	Excellent to good					Fair to poor					

Comparison of Different Soil Classification Systems

Classification System	Based On	Best Use In Highway Engineering
Textural Classification	Particle size	Basic soil texture and drainage behavior
Unified Soil Classification (USCS)	Gradation, plasticity	Geotechnical investigations, embankment stability
BIS Soil Classification	Particle size, plasticity	Indian highway projects, pavement subgrade
HRB/AASHTO Classification	Grain size, plasticity, CBR	Pavement design and highway subgrade evaluation

Evaluation of Soil Strength Tests

Soil strength evaluation is crucial for highway engineering to determine the **load-bearing capacity**, **stability**, and **suitability** of soil for pavement and embankments. The most common soil strength tests include:

1.Direct Shear Test

2.Triaxial Compression Test

3.California Bearing Ratio (CBR) Test

4.Plate Bearing Test

1) Direct Shear Test (IS: 2720-Part 13)

Purpose:

•Determines **shear strength parameters** of soil: **Cohesion (c) and Angle of Internal Friction (ϕ).**

•Used for **sandy and granular soils**.

Apparatus:

•Shear box

•Loading frame

•Proving ring for load measurement

•Dial gauges for displacement



Procedure:

1.The soil sample is placed in the **shear box**.

2.A **normal load (σn)** is applied.

3.A **horizontal shear force** is applied until failure.

4.The test is repeated for different normal loads.

5.The **shear strength (τ)** is calculated from:

Formula:

$$\tau = c + \sigma_n \tan \phi$$

Where:

- τ = Shear strength
- c = Cohesion
- σ_n = Normal stress
- ϕ = Angle of internal friction

2) Triaxial Compression Test (IS: 2720-Part 11)

Purpose:

•Determines **shear strength parameters (c & ϕ)** for cohesive and granular soils.

•Used for **both saturated and unsaturated soils**.

Types of Tests:

1.Unconsolidated Undrained (UU) Test – Quick test, no drainage.

2.Consolidated Undrained (CU) Test – Allows consolidation but no drainage during shearing.

3.Consolidated Drained (CD) Test – Slow test, allowing full drainage.

Apparatus:

•Triaxial cell

•Pressure chamber

•Loading system

•Pore pressure gauge

Procedure:

1.The soil sample is placed inside a **rubber membrane** and subjected to **confining pressure (σ3\sigma_3σ3)**.

2.Axial load (σ1\sigma_1σ1) is applied until failure.

3.Deviator stress and pore pressure readings are recorded.

4.The **Mohr's Circle** is used to determine **cohesion (c)** and **friction angle (ϕ)**.

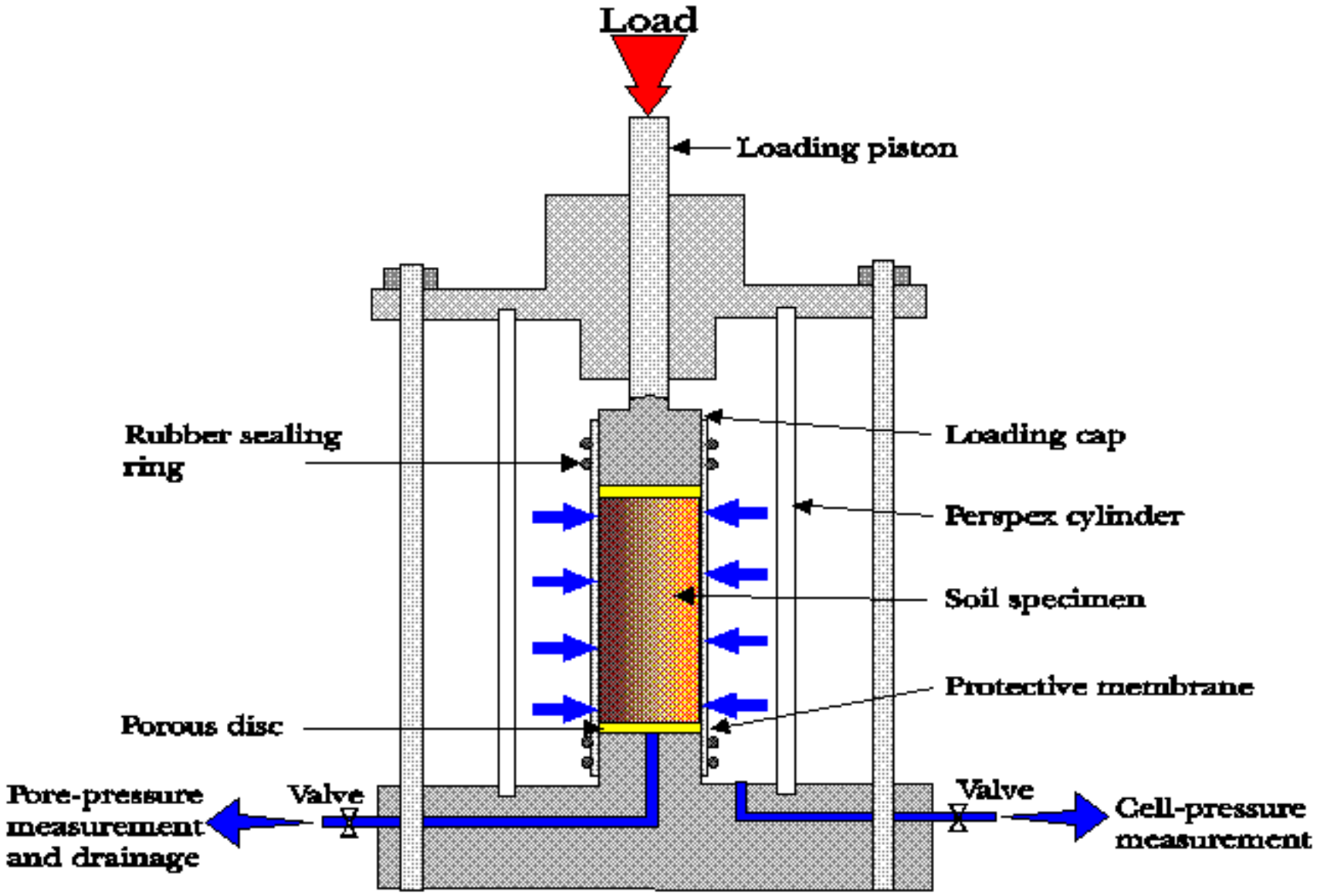
Formula:

$$\sigma_1 = \sigma_3 + \Delta\sigma$$

$$\tau = c + \sigma_n \tan\phi$$

Where:

- σ_1 = Major principal stress
- σ_3 = Minor principal stress
- $\Delta\sigma$ = Deviator stress



Triaxial apparatus

3) California Bearing Ratio (CBR) Test (IS: 2720-Part 16)

- **The California Bearing Ratio (CBR) test was developed by the California State Highway Department as a method for evaluating the strength of subgrade soil and other payment materials for the design and construction of flexible pavements**.
- The 2 test results have been correlated with flexible pavement thickness requirements highways and air fields. Being an empirical test method, CBR test results cannot de related accurately with any fundamental property of the soil or pavement material axed

- The CBR method of test has also been standardized by the Bureau of Indian standards (BIS)
- Procedure for field determination of CBR value of soil in-place or in-situ has also been developed and standardized by different agencies including the BIS.

- The basic principle in CBR test is by causing a cylindrical plunger of 50 mm diameter to penetrate into the specimen of soil or pavement component material at a me of 1.25mm per minute.
- The loads required for 2.5 mm and 5.0 mm penetration of he plunger into the soil/material tested are recorded. The CBR value of the material tested is expressed as a percentage of standard load value in a standard material.
- The standard load values have been established based on a large number of tests on standard crushed stone aggregates at the respective penetration levels of 2.5 and 5.0

3) California Bearing Ratio (CBR) Test (IS: 2720-Part 16)

Purpose:

- Measures soil strength for **subgrade, sub-base, and base course** design.
- Used to determine **pavement thickness**.

Apparatus:

- CBR testing machine
- Standard plunger (50 mm dia)
- Dial gauges
- Load frame

Procedure:

1. The soil sample is **compacted** in a **CBR mold** and soaked for **4 days**.
2. A **50 mm diameter plunger** is pressed into the soil at **1.25 mm/min**.
3. The load is measured at **2.5 mm** and **5 mm** penetration of plunger into soil
4. CBR value is calculated using:

Formula:

$$CBR(\%) = \frac{\text{Load sustained by the specimen at 2.5 or 5mm penetration}}{\text{Load sustained by standard aggregates at corresponding penetration level}} \times 100$$

Penetration Depth (1) mm	Unit Standard Load (2) kg/cm²	Total Standard Load (3) kgf
2·5	70	1 370
5·0	105	2 055

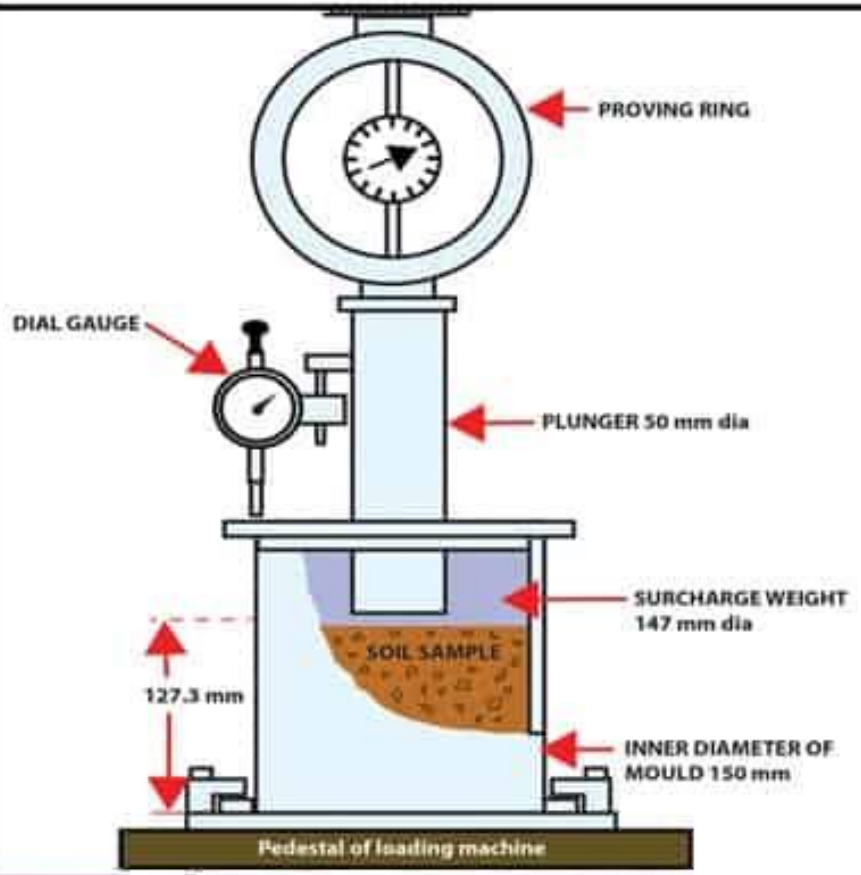
Procedure in detail :

1. The laboratory CBR apparatus consists of a mould 150 mm diameter with a base plate and a collar, a loading frame and dial gauges for measuring the penetration values and the expansion on soaking.
2. The specimen in the mould is soaked in water for four days and the swelling and water absorption values are noted. The surcharge weight is placed on the top of the specimen in the mould and the assembly is placed under the plunger of the loading frame.
3. Load is applied on the sample by a standard plunger with dia of 50 mm at the rate of 1.25 mm/min. A load penetration curve is drawn. The load values on standard crushed stones are 1370 kg and 2055 kg at 2.5 mm and 5.0 mm penetrations respectively.
4. CBR value is expressed as a percentage of the actual load causing the penetrations of 2.5 mm or 5.0 mm to the standard loads mentioned above.
5. Two values of CBR will be obtained. If the value of 2.5 mm is greater than that of 5.0 mm penetration, the former is adopted. If the CBR value obtained from test at 5.0 mm penetration is higher than that at 2.5 mm, then the test is to be repeated for checking.
6. If the check test again gives similar results, then higher value obtained at 5.0 mm penetration is reported as the CBR value. The average CBR value of three test specimens is reported as the CBR value of the sample.

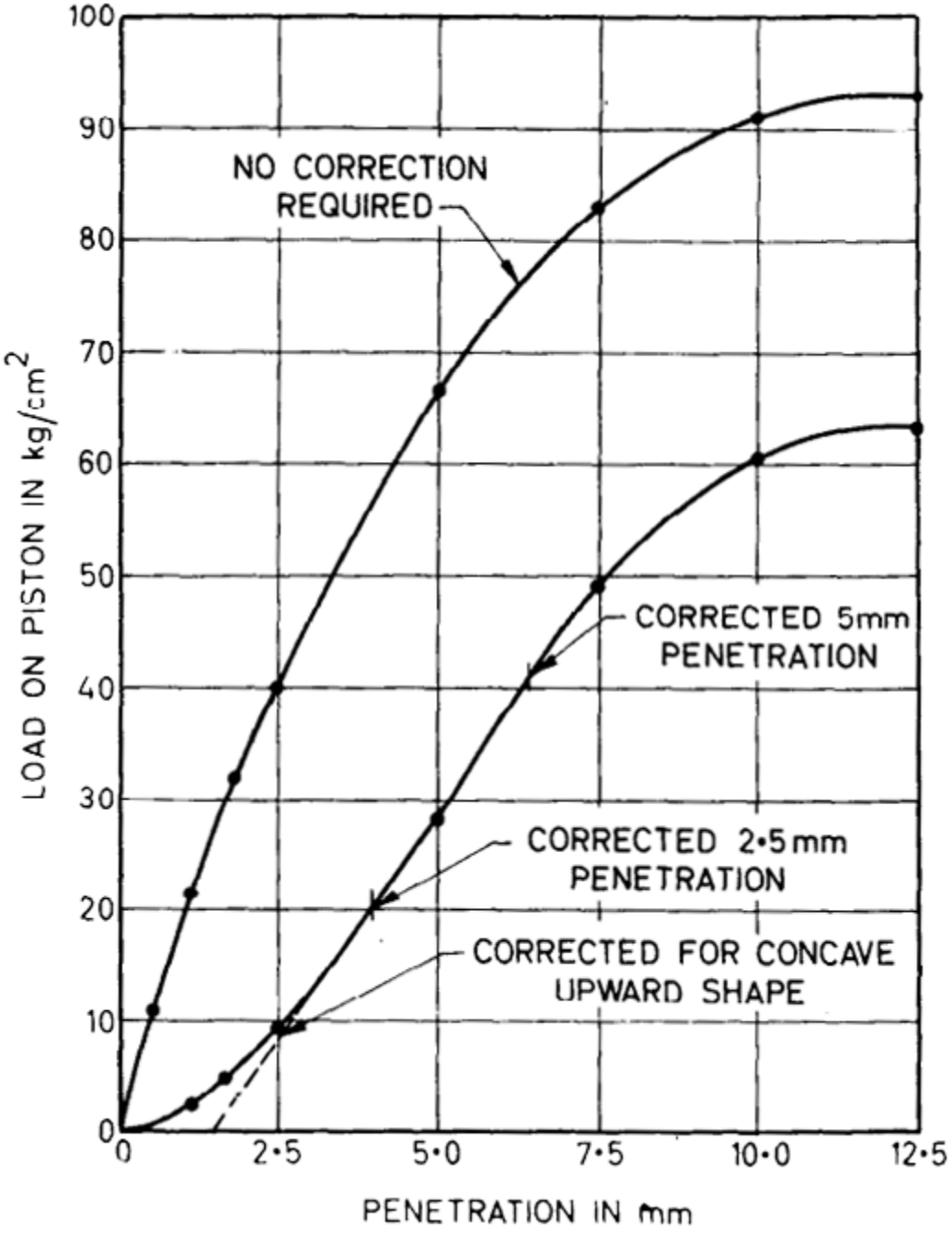
Soil Classification and Typical CBR Values Table

Soil Type / Classification	USCS / IS Classification	Typical CBR Range (%)	Remarks
Well-graded gravel (GW)	Gravel with little fines	40 – 80	Excellent subgrade material
Poorly graded gravel (GP)	Uniform gravel	30 – 60	Good load-bearing capacity
Silty gravel (GM)	Gravel with silt	20 – 40	Slightly reduced due to fines
Clayey gravel (GC)	Gravel with clay	20 – 40	Acceptable for sub-base with compaction
Well-graded sand (SW)	Sand with little fines	30 – 60	Good subgrade material
Poorly graded sand (SP)	Uniform sand	10 – 40	Can be improved with compaction
Silty sand (SM)	Sand with silt	10 – 30	Moderate strength, sensitive to moisture
Clayey sand (SC)	Sand with clay	10 – 30	Better than silty sand
Silt (ML)	Low plasticity silt	5 – 15	Low bearing capacity, moisture-sensitive
Inorganic clay of low plasticity (CL)	Lean clay	5 – 15	Poor to fair, use only with treatment
Silt of high plasticity (MH)	Elastic silt	2 – 8	Very poor subgrade material
Clay of high plasticity (CH)	Fat clay	1 – 5	Very poor; swelling problem; not recommended
Organic silts/clays (OL/OH)	Organic soils	< 2	Unsuitable for pavement support
Peat or highly organic soils (PT)	Highly organic	< 1	Unusable for any engineering purpose




PROVING RING
DIAL GAUGE
PLUNGER 50 mm dia
SURCHARGE WEIGHT
147 mm dia
SOIL SAMPLE
127.3 mm
INNER DIAMETER OF
MOULD 150 mm
Pedestal of loading machine

CBR Test


100
90
80
70
60
50
40
30
20
10
0
LOAD ON PISTON IN kg/cm2
NO CORRECTION
REQUIRED
CORRECTED 5mm
PENETRATION
CORRECTED 2·5 mm
PENETRATION
CORRECTED FOR CONCAVE
UPWARD SHAPE
0
2·5
5·0
7·5
10·0
12·5
PENETRATION IN mm

4) Plate Bearing Test (IS: 1888-1982)

Purpose:

•Determines **bearing capacity** and **modulus of subgrade reaction (k-value).**

•Used for **highway embankments, airport pavements, and foundations**.

Apparatus:

•**Steel plate (300–750 mm dia)**

•Hydraulic jack

•Dial gauges

•Load cell

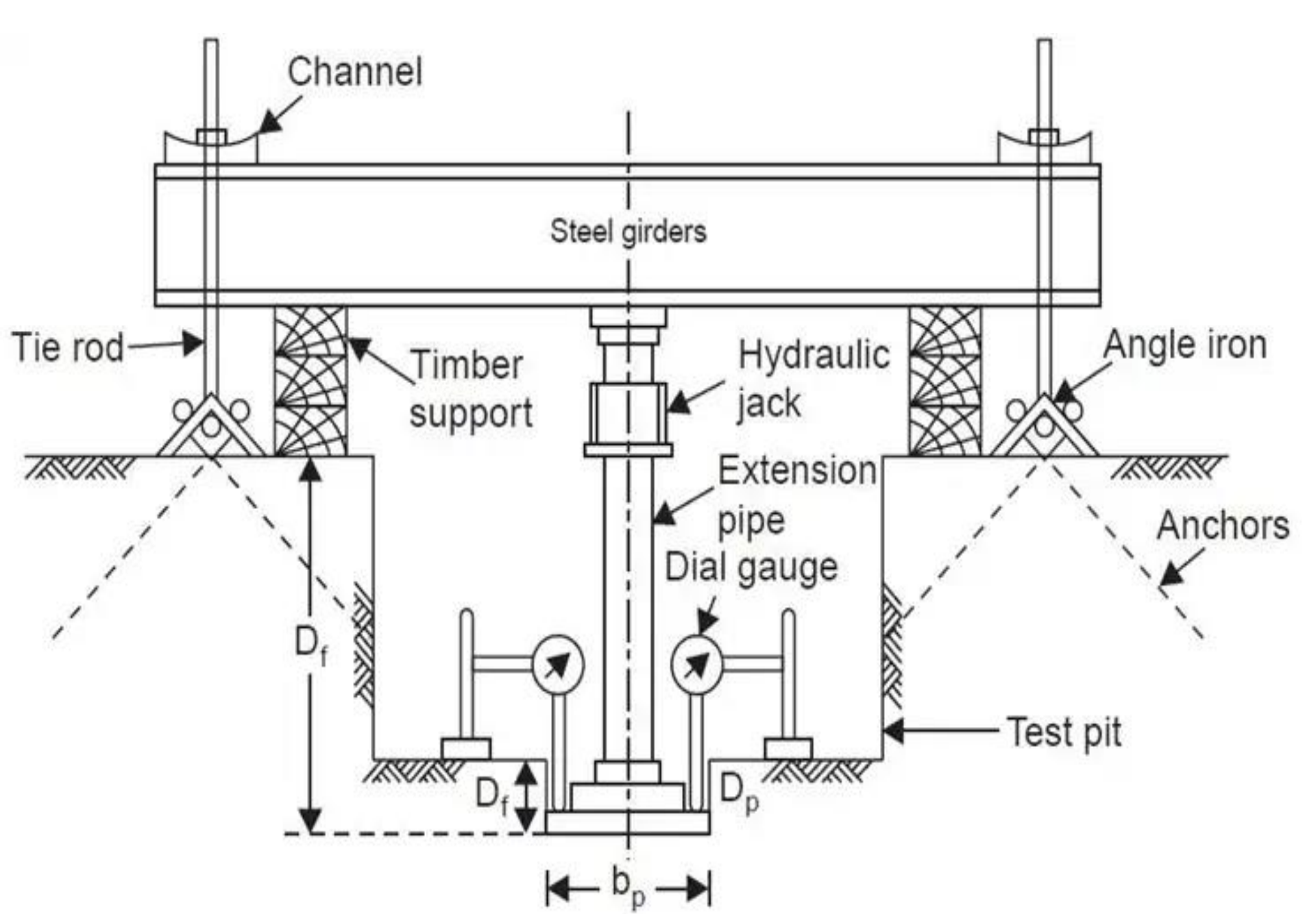
Procedure:

1.A **rigid steel plate** is placed on the **soil surface**.

2.A **gradual load is applied** using a hydraulic jack.

3.The **settlement is recorded** at different loads.

4.Load-Settlement curves are plotted.


Channel
Steel girders
Tie rod
Timber support
Hydraulic jack
Angle iron
Extension pipe
Anchors
Dial gauge
D_f
Test pit
D_f
D_p
b_p

Formula:

$$K = \frac{P}{\Delta}$$

Where:

- K = Modulus of subgrade reaction
- P = Applied pressure
- Δ = Settlement

📌 **Relationship with CBR:**

$$K = 1.5 \times \text{CBR}$$

Typical K-Values:

Soil Type	K-Value (kg/cm²/cm)
Soft Clay	< 2.5
Medium Clay	2.5 - 5.0
Sand	5 - 10
Gravel	> 10

Desirable Properties of Aggregates:-

1) Strength, 2) Toughness, 3) Hardness,

4) Shape, 5) Adhesion with Bitumen, 6) Durability &

7) Free from Deleterious Material.

Strength :- (Crushing Test)

- The aggregates used in top layers are subjected to :
  - (i) Stress action due to traffic wheel load.
  - (ii) Wear and tear.
  - (iii) Crushing.
- For a high quality pavement, the aggregates should posses high resistance to crushing, wear & tear. Aggregates also withstand the stresses due to traffic wheel load.

Toughness :- (Impact Test)

1) Resistance of the aggregates to impact is termed as toughness.
2) Aggregates used in the pavement should be able to resist the effect caused by impact due to the jumping of the steel tyre wheels from one particle to another.

Hardness :- (Abrasion Test)

1) The aggregates used in the surface course are subjected to constant rubbing or abrasion due to moving traffic.
2) The aggregates should be hard enough to resist the abrasive action caused by the movements of traffic.
3) The abrasive action is severe when steel tyred vehicles moves over the aggregates exposed at the top surface.

Shape of Aggregates :- (Shape Test)

1) Aggregates which happen to fall in a particular size range may have rounded, cubical, angular, flaky or elongated particles.
2) The flaky & elongated particles will have less strength & durability when compared with cubical, angular or rounded particles of the same aggregate.

Adhesion with Bitumen :- (Bitumen Adhesion Test)

1) The aggregates used in bituminous pavements should have less affinity with water when compared with bituminous materials, otherwise the bituminous coating on the aggregate will be stripped off in presence of water.

Durability :- (Soundness Test)

1) The property of aggregates to withstand adverse action of weather is called soundness.
2) The aggregates are subjected to the physical and chemical action, hence it is desirable that the aggregates used should be sound enough to withstand the weathering action .

Test on Aggregate :-

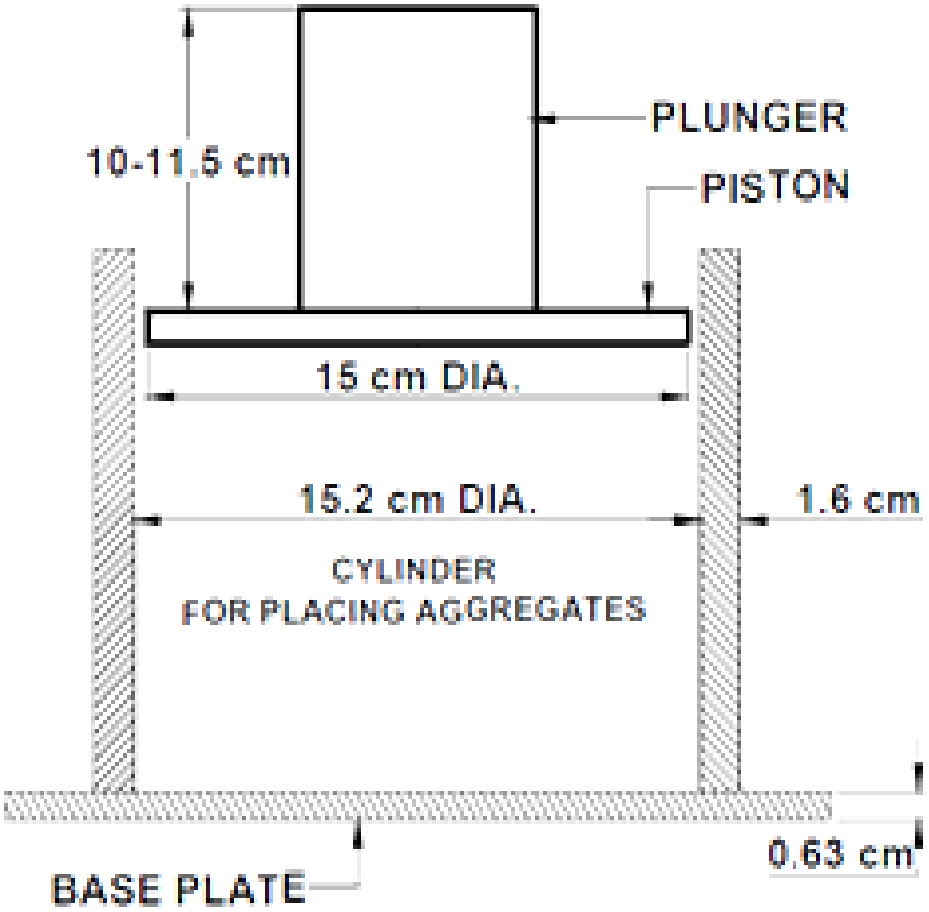
The suitability of aggregate for pavement construction, is checked by following tests:

1) Crushing Test
2) Abrasion Test
3) Impact Test
4) Soundness Test
5) Shape Test
6) Specific Gravity & Water Absorption Test
7) Bitumen Adhesion Test

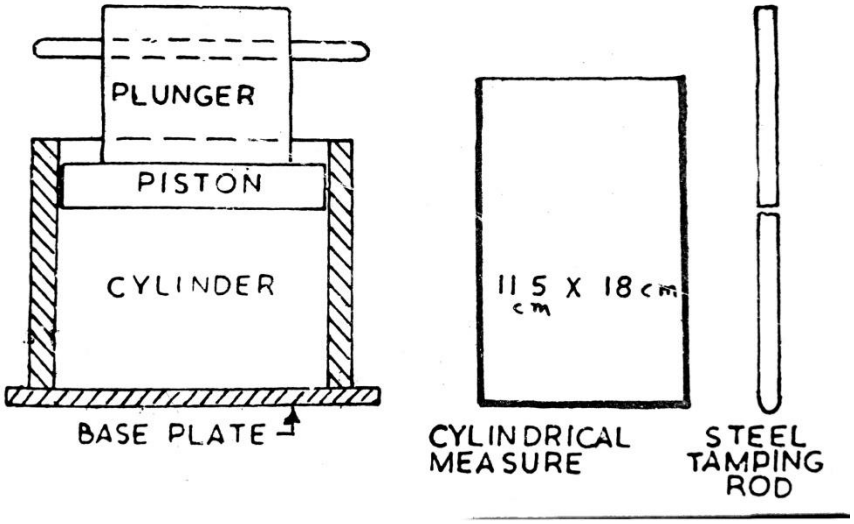
Crushing Test

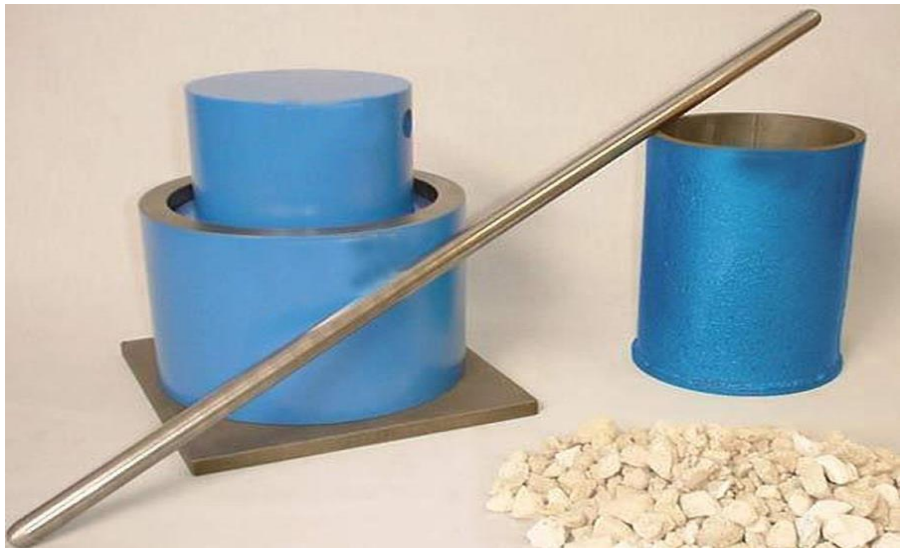
1) The aggregate crushing value provides a relative measure of resistance to crushing under gradually applied crushing load.
2) The test consists of the specimen of aggregate in standard mould to a compression test under standard load conditions.
3) Dry aggregates passing through 12.5 mm sieves and retained on 10 mm sieves are filled in a cylindrical measure in three layers, each layer is tampered 25 times with at standard tamping rod. The test sample is weighed (W1 in kg)
4) The plunger is placed on the top of specimen and a load of 40 tones is applied at a rate of 4 tones per minute by the compression machine.
5) The crushed aggregate is removed and sieved on 2.36 mm IS sieve, the crushed material which passes this sieve is weight (W2 in kg).
6) The aggregate crushing value is the percentage of the crushed material passing 2.36 mm IS sieve (W2) in terms of original weight of the specimen. (W1)

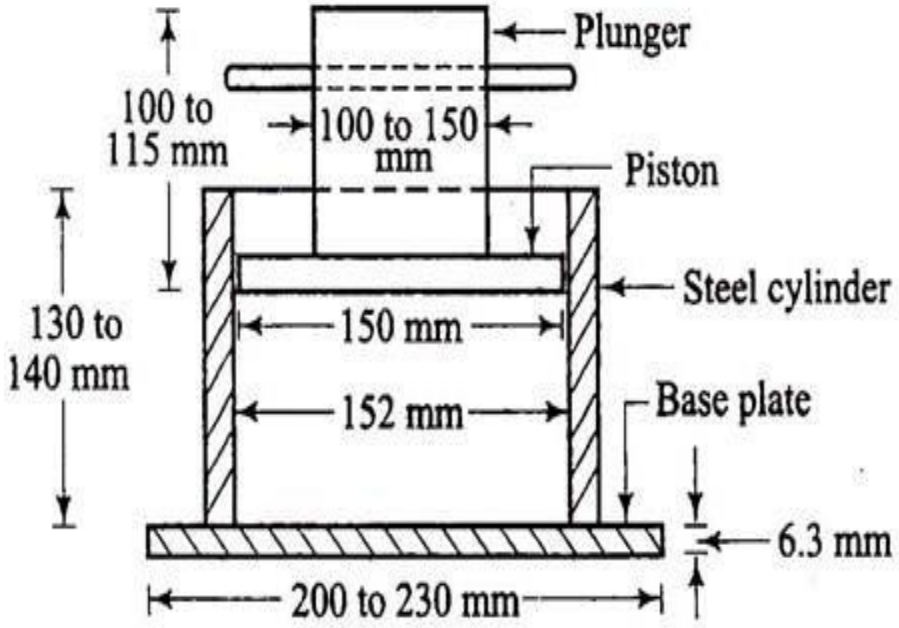
- **Strong** aggregate give **low** crushing value.
- The aggregate crushing value for good quality aggregate to be used in **base course shall not exceed 45 %** and the value for **surface course shall be less than 30 %.**
- Aggregates Crushing Value = [(W2) / (W1)] * 100



Crushing test Apparatus







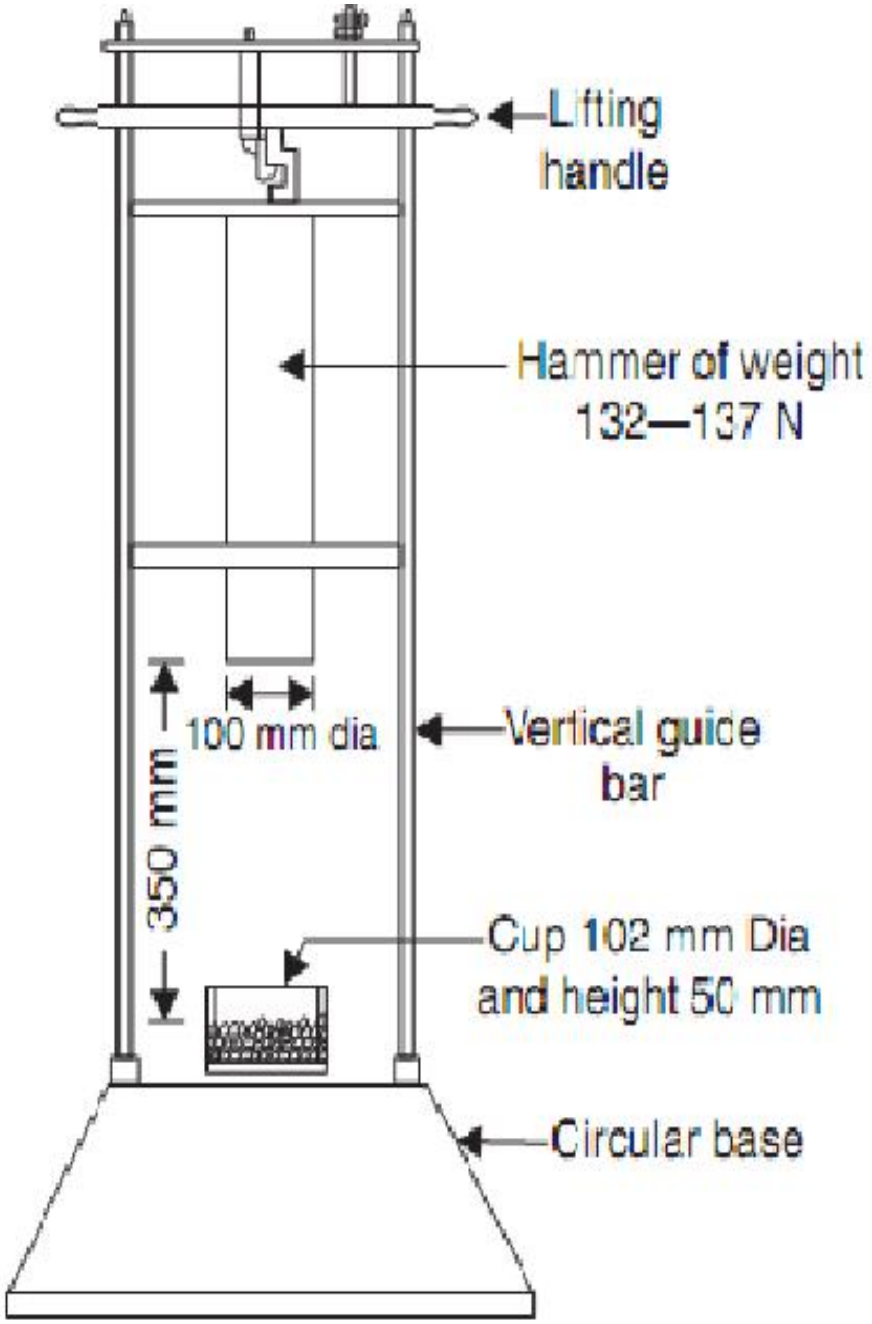
Aggregate crushing test apparatus



Impact Test:-

1) The aggregate impact test is carried out to evaluate the resistance to impact of aggregates.
2) Aggregates passing 12.5 mm sieve& retained on 10 mm sieve is filled in a cylindrical steel cup of internal diameter 10.2 cm and depth 5 cm which is attached to a metal base of impact testing machine.
3) The material is filled in three layers and each layer is tamped by giving 25 number of blows.
4) Metal hammer of weight 13.5 to 14 Kg is arranged to drop with a free fall of 38.0 cm by vertical guides and the test specimen is subjected to 15 number of blows.
5) The crushed aggregate is allowed to pass through 2.36 mm IS sieve. The aggregate impact value is expressed as the percentage of the fine formed in terms of the total weight of the sample.
6) Aggregates to be used for wearing course, the impact value shouldn't exceed 30 %. For bituminous macadam the maximum permissible value is 35%. For Water bound macadam base courses the maximum permissible value defined by IRC is 40%.




Lifting handle
Hammer of weight 132—137 N
100 mm dia
350 mm
Vertical guide bar
Cup 102 mm Dia and height 50 mm
Circular base

Mechanical Prosperities of Aggregate : Aggregate Impact Value

- The crushed aggregate is removed from the cup and the whole of it is sieved on 2.36 mm I.S. Sieve.
- $The\ Aggregate\ Impact\ Value = \frac{B}{A} \times 100$

  B = weight of fraction passing 2.36 mm I.S. Sieve.

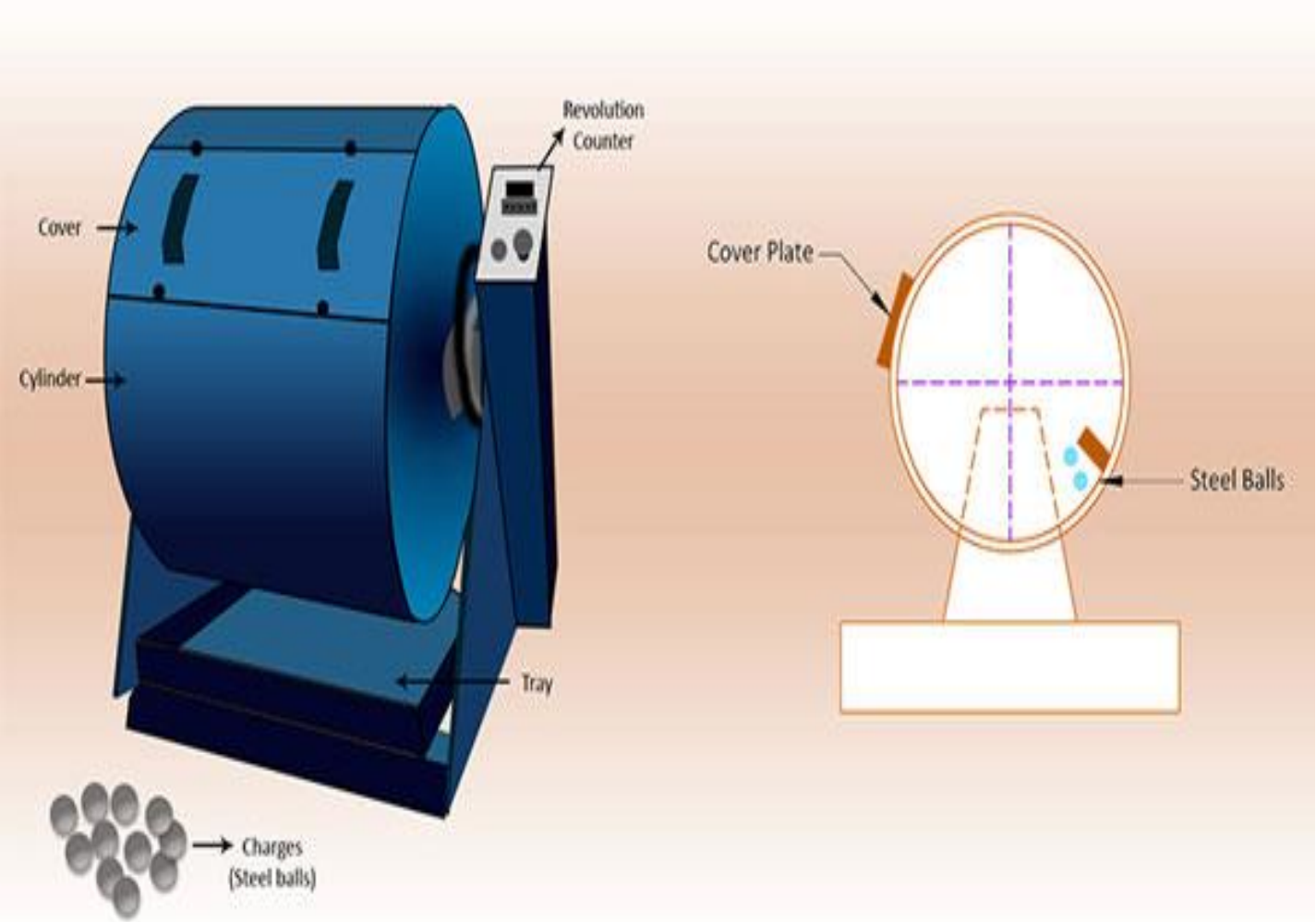
  A = weight of oven-dried sample.
- The aggregate impact value should not be more than 45 per cent by weight for aggregates used for concrete other than wearing surfaces and 30 per cent by weight for concrete to be used as wearing surfaces, such as runways, roads and pavements.

Abrasion Test: - Los Angeles Test

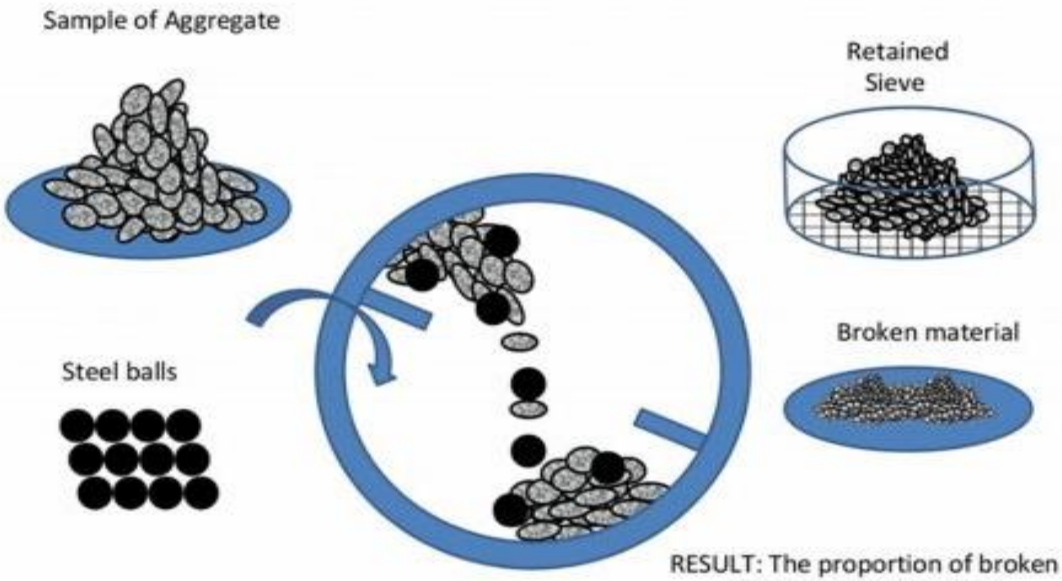
1) Abrasion test is carried out to test the hardness property of aggregates. (to decide whether they are suitable for pavement construction)
2) Los Angeles abrasion test is a preferred in India (IS:2386 part-IV).
3) Los Angeles abrasion test is to find the percentage wear due to relative rubbing action between the aggregate & steel balls used as abrasive charge.
4) Los Angeles machine consists of circular drum of internal diameter 70cm & length 50cm mounted on horizontal axis enabling it to be rotated.
5) An abrasive charge consisting of cast iron spherical balls of weight 390-445 g is placed in the cylinder along with the 5-10 kg aggregates.
6) The cylinder is then locked & rotated at the speed of 30-33 rpm for a total of 500 - 1000 revolutions depending upon the gradation of aggregates.
7) After specified revolutions, the material is sieved through 1.7 mm IS sieve and the weight of powdered aggregate passing this sieve is found.

Abrasion Test: - Los Angeles Test

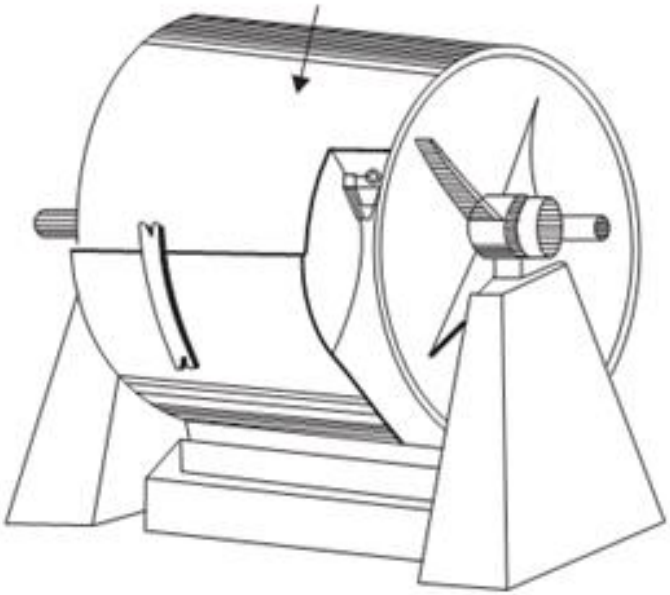
1) Thus the abrasion value will the percentage wear or the percentage passing 1.7 mm sieve expressed in terms of the original weight of the sample.
2) A value upto 50% are allowed in base course like water bound and bituminous macadam

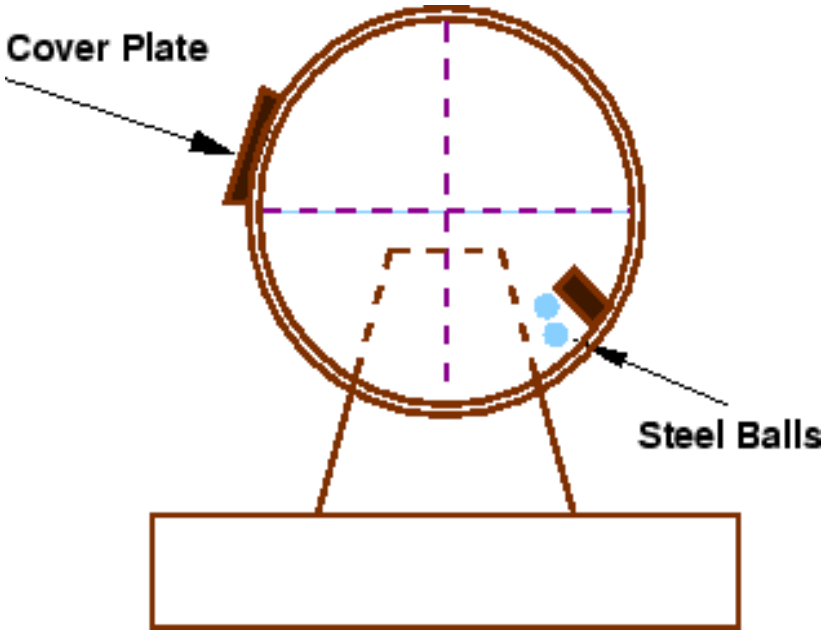


Los Angeles Test



RESULT: The proportion of broken material, expressed as a percentage





Mechanical Prosperities of Aggregate : Aggregate Abrasion Value

Table 3.22. Gradings of Test Samples

Sieve Size		Weight in gm. of Test Sample For Grade						
Passing	Retained on	A	B	C	D	E	F	G
mm	mm							
80	63	-	-	-	-	2500	-	-
63	50	-	-	-	-	2500	-	-
50	40	-	-	-	-	5000	5000	-
40	25	1250	-	-	-	-	5000	5000
25	20	1250	-	-	-	-	-	5000
20	12.5	1250	2500	-	-	-	-	-
12.5	10	1250	2500	-	-	-	-	-
10	6.3	-	-	2500	-	-	-	-
6.3	4.75	-	-	2500	-	-	-	-
4.75	2.36	-	-	-	5000	-	-	-

Mechanical Prosperities of Aggregate : Aggregate Abrasion Value

Table 3.21. Specified Abrasive Charge

Grading	*Number of spheres*	*Weight of charge (gm)*
A	12	5000 ± 25
B	11	4584 ± 25
C	8	3330 ± 20
D	6	2500 ± 15
E	12	5000 ± 25
F	12	5000 ± 25
G	12	5000 ± 25

Soundness Test:-

1) Soundness test is intended to study the **resistance of aggregates to weathering action**, by conducting accelerated weathering test cycles.
2) The porous aggregates subjected to **freezing and thawing** are likely to disintegrate prematurely.
3) To ascertain the durability of such aggregates, they are subjected to an accelerated soundness test as specified in IS:2386 part-V.
4) Aggregates (specified size) are subjected to cycles of alternate wetting in a saturated solution of either **sodium sulphate or magnesium sulphate** for 16 - 18 hours and then dried at 110^0C in oven (constant weight)
5) After five cycles, **the loss in weight of aggregates** is determined by sieving out all undersized particles and weighing.
6) And the loss in weight should not exceed 12 percent when tested with sodium sulphate and 18 % with magnesium sulphate solution.

Soundness Test



Before



After





Bitumen Adhesion Test :-

1) **Bitumen adheres well** to all normal types of road aggregates provided they are **dry and free from dust.**
2) In the absence of water there is practically no adhesion problem of bituminous construction.
3) **Adhesion problem** occurs when the aggregate is **wet and cold.**
4) This problem can be dealt with by removing moisture from the aggregate **by drying and increasing the mixing temperature.**
5) Several laboratory tests are conducted to arbitrarily determine the adhesion of bitumen binder to an aggregate in the presence of water.
6) **Static immersion test** is one specified by IRC and is quite simple. The principle of the test is by immersing **aggregate fully coated with binder in water** maintained at temperature **40^0C for 24 hours**.
7) IRC has specified maximum **stripping value** of aggregates should **not exceed 5%.**

Shape Test :-

1) Shape test pertain to the **shape of particles.**
   - **I. Flakiness index,**
   - **II. Elongation index &**
   - **III. Angularity number.**

2) The flakiness index is the **% by weight of aggregate** whose **least dimension is less than 0.6 of their mean dimension**. This index should be less than 15%.

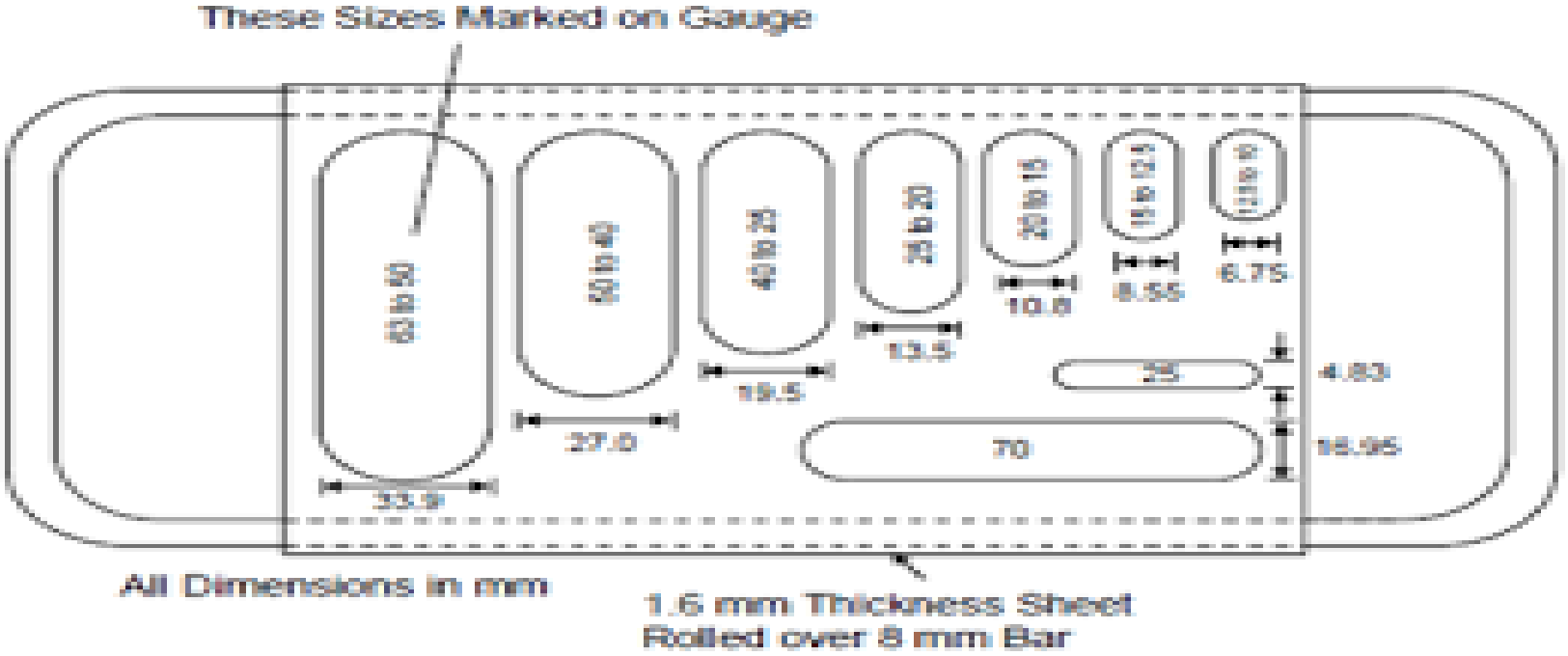
3) Elongated aggregates having length greater than the specified gauge (**ie length greater than 1.8 times their mean dimension**) each size range is weighed and this weight of stones is expressed as % of total weight of sample to get elongation index. This index should be less than 15%.

Flakiness Index :-

1) It is a **% by weight of aggregates particles whose least dimension is less than 0.6 of their mean dimension.**
2) Sufficient aggregate sample is sieved through a set of sieves & separated into specified size ranges.
3) To separate the flaky material, the aggregates which pass through the appropriate slot of the standard thickness gauge are found.
4) The flaky material passing through appropriate slot from each size range of test aggregates are **added up** & let this be weight '**w**' if the total weight of sample taken from different size ranges is '**W**'
5) The flakiness index is given by = (w /W) * 100
6) Flakiness index is less than 15% & does not exceed 25%.

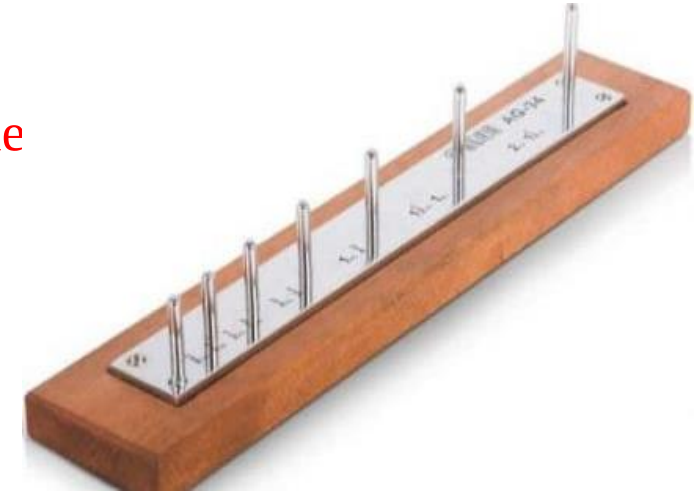
Thickness Gauge





Elongation Index :-

1) The elongation index of an aggregates is the percentage by weight of particles whose greatest dimension on length is greater than 1.8 times their mean dimension.
2) Total weight of the elongation stones (retained) is expressed as % of total weight of sample to get the elongation index.
3) Elongation index value in excess of 15% is undesirable



Elongation Gauge

as per IS: 2386 (Part 1)



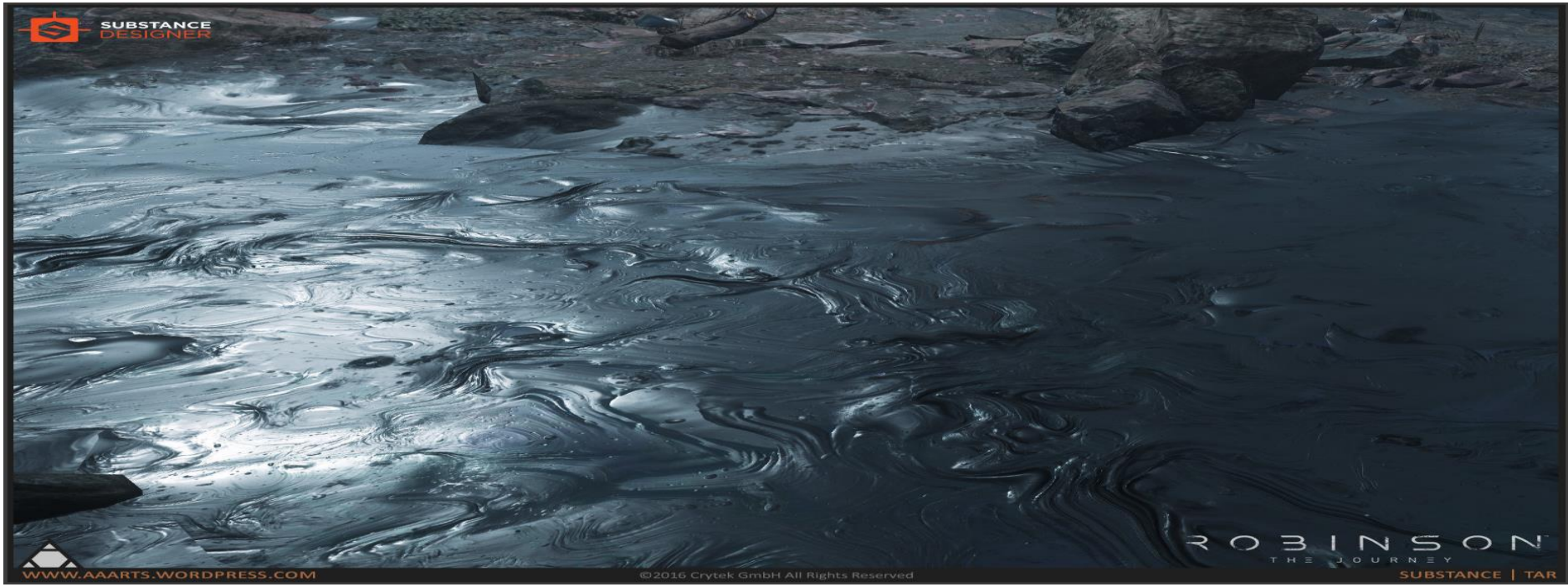
- **Bitumen** a **hydro-carboneous material** of either natural or pyrogenous origin, found in gaseous, liquid, semi-solid in state & completely soluble in carbon disulphide is called bitumen.

- **Bitumen** is a **petroleum product obtained by distillation (**the action of purifying a liquid by a process of heating and cooling) of **petroleum crude.** Bituminous material are used in road construction because of their binding & water proofing properties.

- **Different Forms of Bitumen:**
    1) Cutback Bitumen
    2) Bitumen Emulsion
    3) Bitumen Primers
    4) Modified Bitumen






SUBSTANCE
DESIGNER
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THE JOURNEY
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SUBSTANCE | TAR

Cutback Bitumen:

1) Normal practice is to **heat bitumen to reduce its viscosity**. **In some situations preference is given to use liquid binders such as cutback bitumen.**
2) In cutback bitumen, suitable solvent (volatile diluent) is **used to lower the viscosity of the bitumen.**
3) **Used in cold weather**. Kerosene, naphtha, furnace oil distillates are used for preparation of cutback bitumen.
4) Types: Rapid Curing, Medium Curing, Slow Curing.

Bitumen Emulsion:

1) It is a **liquid product in which bitumen is suspended in a finely divided condition** in an aqueous medium and stabilized by suitable material.
2) **Emulsion** is a liquid product obtained by vigorously stirring up a mixture of two unmixable liquids is known as emulsion.

Bituminous Primers :

1) In bituminous primer the **distillate is absorbed by the road surface on which it is spread.**
2) Bitumen primers are **useful on the stabilized surface & water bound macadam base courses**. Bituminous primers are generally prepared on road sites by mixing penetration bitumen with petroleum distillate.

Modified Bitumen :

1) **Certain additives or blend of additives called as bitumen modifiers** can improve properties of bitumen & bitumen mixes.
2) Bitumen treated with these modifiers is known as modified bitumen.
3) Polymer Modified Bitumen & Crumb Rubber Modified Bitumen.

Forms of Bitumen and Their Uses (As per IRC Guidelines)

Form of Bitumen	Description	Types / Grades	Typical Uses	IRC Reference
Cutback Bitumen	Bitumen diluted with volatile solvents to reduce viscosity for ease of application at low temperatures.	RC (Rapid Curing), MC (Medium Curing), SC (Slow Curing)	**Used in cold climate regions, patch repair, and prime coats.**	IRC: 120, IRC: SP:53
Bitumen Emulsion	Bitumen suspended in water with emulsifying agents, suitable for cold applications.	Cationic (RS, MS, SS), Anionic (less common)	**Surface dressing, tack coat, patching, slurry seal, fog seal.**	IRC: SP:100, IRC: 120
Bitumen Primer	Low-viscosity bituminous liquid applied to prepare and seal the surface before laying bituminous layers.	-	**Applied on granular bases (WBM/WMM) before bituminous layers.**	IRC: 16, IRC: SP:100
Modified Bitumen	Bitumen modified with polymers (elastomers, plastomers) or rubber for enhanced performance.	PMB (Polymer Modified Bitumen), CRMB (Crumb Rubber Modified Bitumen)	**Heavy traffic roads, highways, bridge decks, areas with temperature extremes.**	IRC: SP:53, IRC: SP:107, MoRTH 2013

Tar is **viscous liquid which is produced by destructive distillation of organic matter such as coal or wood**. Based on the material, it is referred to coal tar or wood tar. There are five grades of tar for roads i.e. RT-1 to RT-5. **No longer used for road construction due to carcinogenic effect.**

The carcinogenic effect of bitumen is a concern primarily due to **exposure to bitumen fumes and polycyclic aromatic hydrocarbons (PAHs) during heating and handling**, especially in paving and roofing operations

Bitumen is obtained refining process (Fractional Distillation) & Tar is obtained by complete distillation process.

Asphalt (India & Europe) = Aggregates + Bitumen i.e. Paving Material

Asphalt (America) = Bitumen (Black Viscous Material)

Requirements of Bitumen:

The desirable properties of bitumen **depend on the mix type & construction.** In general, bitumen should posses following properties:

1) The bitumen should not be highly **temperature susceptible**. During summer should not become to soft or unstable & during winter should not become too brittle causing cracks.
2) The **viscosity** of the bitumen at the time of mixing & compaction should be **adequate.**
3) Bitumen should bond properly with the aggregates so they hold together firmly in the mix.

Difference Between Tar and Bitumen

Aspect	Tar	Bitumen
Source	By-product of destructive distillation of coal or wood	By-product of refining crude petroleum oil
Appearance	Dark brown to black, more viscous and sticky	Black, thick, but smoother in texture when hot
Odour	Has a strong, unpleasant smoky smell	Mild petroleum smell when heated
Volatility	More volatile (produces more fumes on heating)	Less volatile, safer to handle at high temperatures
Toxicity	Contains higher carcinogenic compounds (PAHs)	Less toxic, but still requires safety precautions
Adhesion to Aggregates	Generally poorer than bitumen	Good natural affinity to aggregates
Water Resistance	Less water-resistant	Better water resistance
Uses (Past and Present)	Previously used in road construction, waterproofing; now rarely used	Widely used in roads, roofing, and waterproofing
Modern Status	Obsolete in road construction due to health and performance issues	Standard binder in flexible pavements

Comparison of Bituminous Materials (As per IRC)

Material	Source	State at Ambient Temp.	Usage	Remarks / IRC References
a) Bitumen	Petroleum refinery by-product (from crude oil)	Solid / Semi-solid	Binder in road construction (flexible pavements)	IRC: 16, MoRTH 500, IS 73
b) Asphalt	Natural or petroleum origin – bitumen + mineral filler	Solid / Semi-solid	Used in hot mix asphalt, surface courses, overlays	Sometimes used synonymously with bitumen in India
c) Tar	By-product of coal/wood destructive distillation	Solid / Semi-solid	Was used in early road works; now obsolete due to toxicity	Not recommended in modern road works (IRC: SP:100)
d) Emulsion	Bitumen + water + emulsifier	Liquid	Cold mix, tack coat, fog seal, slurry seal	IRC: SP:100, IS 8887
e) Cutback	Bitumen + volatile solvent (e.g., kerosene)	Liquid	Prime coat, patching, cold weather work	IRC: SP:53, IS 217
f) Road Oil	Lightly processed petroleum oils or bitumen blends	Liquid	Dust control on unpaved roads, light surface treatments	Mostly used for rural/low-volume roads (not common in India)

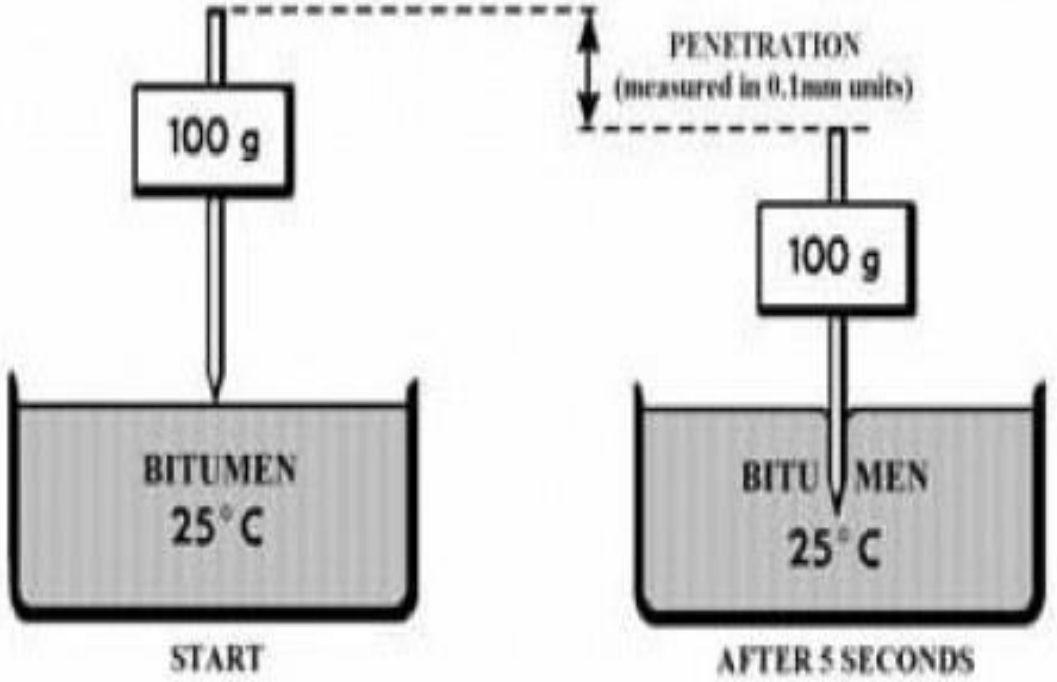
Test for Bituminous Material

1) Penetration Test
2) Ductility Test
3) Softening Point Test
4) Flash & Fire Point Test
5) Specific Gravity Test
6) Viscosity Test
7) Float Test
8) Loss on Heating Test
9) Water Content Test
10) Solubility Test
11) Spot Test

Penetration Test: Evaluation of Grade of Bitumen

1) This test determines **the hardness or softness of bitumen**.
2) This test is done with the help of **penetrometer** which consists of a needle assembly with a total weight of 100g & device releasing & locking in any position.
3) There is a gradual dial to read penetration values to $1/10^{th}$ of a millimeter.
4) Softened bitumen stirred thoroughly & poured into container to a depth at least 15mm in excess of expected penetration.
5) The sampler container are then placed in a temperature controlled water bath at temp. of 25^0C for one hour.
6) The sample with container is taken out & the needle is arranged to make contact with the surface of sample. The dial is set to zero or the initial reading is taken & needle is released for 5 seconds. The final reading is taken on dial gauge.
7) At least three penetration tests are made on this sample by testing at distance of at least 10 mm apart. The mean value is reported as a penetration value.



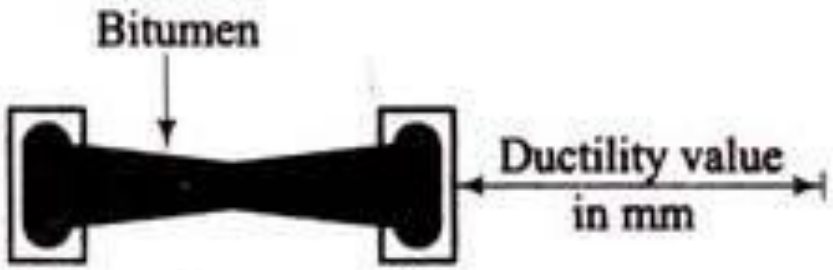

PENETRATION
(measured in 0.1mm units)
100 g
BITUMEN
25°C
START
100 g
BITUMEN
25°C
AFTER 5 SECONDS

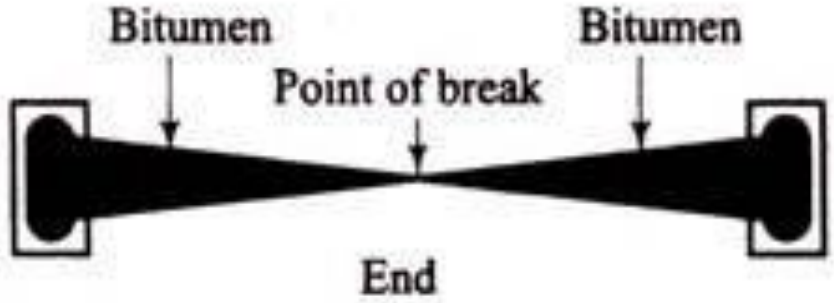
Ductility Test:

1) Ductility is the property of bitumen that permits great **deformation or elongation.**
2) Ductility is the distance in cm, to which a *standard* ***sample*** *or briquette* of the material will be **elongated without breaking**.
3) **Less ductile** bitumen responsible for **cracks** in road surface.
4) Ductility Mould, Heating Equipment, Water Bath, Ductility Testing Machine, Knife, Glyserin & Dextrine.
5) The bitumen sample is **heated** (75–100^{0}C) & **poured** in the mould assembly.
6) These samples with moulds are **cooled** (30-70 minutes) & then kept in **water bath** at 25–27^{0}C temperature (30-70 minutes)
7) The excess bitumen is cut & the surface is **leveled** using a **hot knife**.
8) Samples are **kept in ductility machine** (70-90 minutes) at 25–27^{0}C temperature.
9) **Side pieces** of the moulds are removed, the **clips are hooked** on the machine & the machine is operated at **uniform speed of 5 cm/minute**.

Ductility Test:

1) The **distance up to the point of breaking of thread** is the ductility value which is reported in cm. Ductility value should not exceed ± 10% of mean value.
2) The ductility value gets affected by pouring temp., test temp., rate of pulling, etc.
3) The **minimum ductility value of 75 cm** has been specified by the BIS.
4) Sample should be kept **25 mm below & above in water** in the ductility testing machine.
5) Take the average ductility value of **3** samples.





Ductility test



Bitumen Ductility Test

ASTM D113



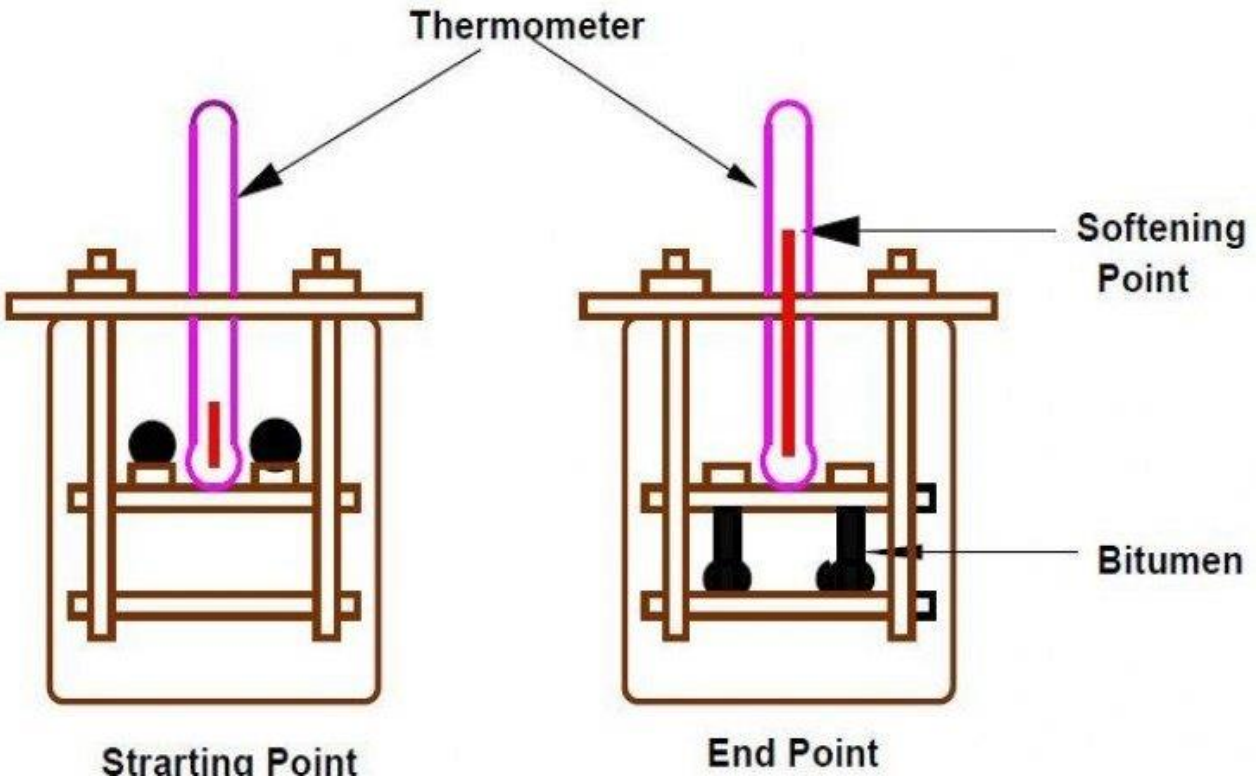


Softening Point Test:

1) Softening points denotes the **temperature** at which the bitumen attains softening or a particular **degree of softening** under the specification of test.
2) **Ring & Ball apparatus,** Beaker, Thermometer, Ball Centering Guides, Ring Holeder Assembly, Heating Equipment, Glycerin, Dextrin, Steel Balls & Rings.
3) Bitumen sample is melted at temp. of (75–100^{0}C) & **poured** in the ring.
4) Allowed it to **cool in the** air for **30 minutes** then surface is leveled by knife.
5) Steel Balls are kept in distilled water at temp. of 5^{0}C for 15 minutes.
6) A **brass ring** containing test sample of **bitumen** is **suspended** in liquid like water or glycerin.
7) A **steel ball** is placed upon the bitumen sample & the liquid medium is **heated** at rate of **5^{0}C per minute** by control heating unit. Stirring of liquid is also done while heating.

Softening Point Test:

1) **Temp. is noted** when the **softened bitumen touches the metal plate** which is at specified distance below.
2) **Average of two** reading (two steel ball) is taken softening point.
3) Any softening value of two value should not be differed by **1^0C** from mean value.
4) **Higher point** indicates **lower temp. susceptibility** & bitumen is preferred in hot climate.




ISOTEX
MAGNETIC STIRRER









Flash & Fire Point Test:

1) **Softening Point** denotes the temperature at which the bitumen attains a particular degree of softening under the specifications of test.
2) **Flash Point** is defined as the lowest temperature at which the vapor of bitumen momentarily catches fire in the form of flash under specified test conditions.
3) **Fire Point is** defined as the lowest temperature under specified test conditions at which the bituminous material gets ignited and burns.
4) The test on bitumen from **safety point of view** during road construction pertains to flash & fire point test of bitumen. Bitumen material leaves out volatiles at some particular temperatures & volatiles catch fire causing a flash.
5) This test is performed **to take precaution against fire hazards** during heating of bitumen.

In simple words:

- **Flash Point** = *Warning sign* (momentary fire ⚡)
- **Fire Point** = *Danger zone* (continuous fire 🔥)

Aspect	Flash Point	Fire Point
Definition	The lowest temperature at which vapors above the bitumen momentarily ignite when a small flame is applied.	The lowest temperature at which vapors ignite and continue to burn for at least 5 seconds.
Purpose	Indicates the temperature at which the material becomes a fire hazard.	Indicates the temperature at which sustained burning can occur.
Observation	Small, brief flash of flame on the surface.	Continuous, steady flame on the surface.
Safety Implication	Alerts about fire risk starting point.	Critical for safe handling at high temperatures.
Test Equipment	Pensky-Martens Apparatus	Pensky-Martens Apparatus
Typical Values	Lower than Fire Point (e.g., Flash Point ≈ 220°C)	Higher than Flash Point (e.g., Fire Point ≈ 240°C)

Flash & Fire Point Test:

1) **Pensky-Martens** closed cup apparatus or open cup are used for the test.
2) Bitumen is kept in oven for melting.
3) After that filled in the cup or lid, up-to a mark.
4) The lid is placed to close the cup in closed system.
5) **Thermometer** of specified range are suitably fixed.
6) The sample is **heated at 5 to 6^0C / minute** with stirring at 60 rpm.
7) The test flame is applied at intervals depending on expected flash & fire points.
8) First application is made at least **17^0C below** the actual flash point (175- 217^0C) & then at every increase **1 to 3^0C.**
9) Before application of flame **stirring is discontinued**.
10) The **flash point** is taken as the temperature read on the thermometer at the time of flame application that causes a **bright flash** in the interior of the cup in closed system.

Flash & Fire Point Test:

1) The heating is continued until the material gets ignited & **continues to burn for 5 seconds**. This temperature is recorded at the **fire point**.
2) The minimum specified flash point of bitumen used in road construction in this closed type test is **175^0C.**



Flash & Fire Point

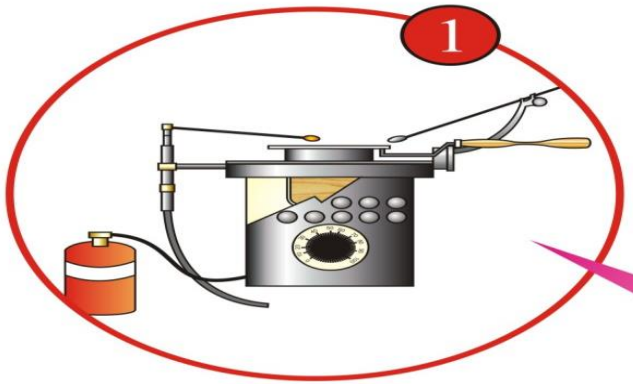
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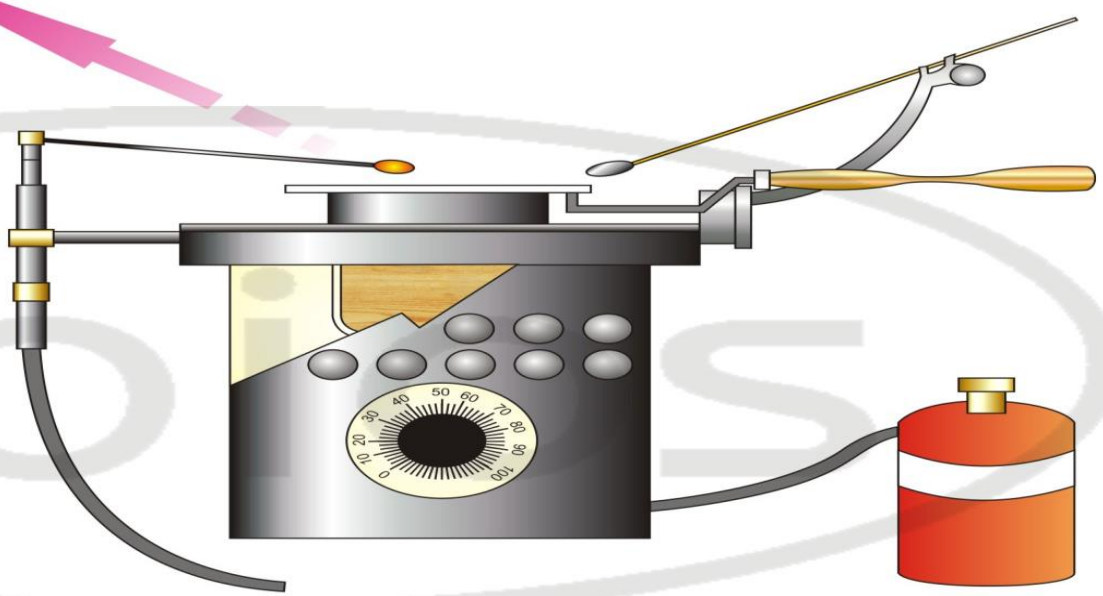
FLASH & FIRE POINT APPARATUS



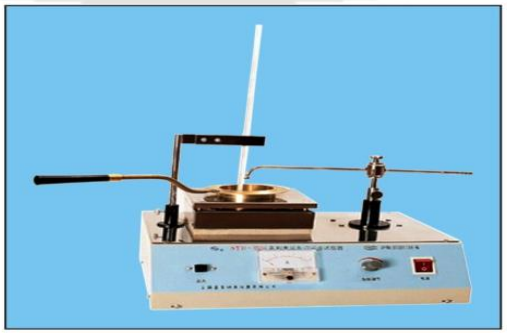
CLEVELAND OPEN CUP

OIL SAMPLE IS POURED INTO CUP. THERMOMETER IS PLACED IN THE OIL AND THE OIL IS HEATED. THE TEST FLAME IS PASSED OVER THE CUP AT EVERY 2ºC INCREASE IN SAMPLE TEMPERATURE. WHEN THE SAMPLE VAPOURS IGNITE MOMENTARILY IN AIR, THE FLASH POINT TEMPERATURE IS REACHED. WHEN THE SAMPLE VAPOURS SUSTAIN COMBUSTION FOR AT LEAST FIVE SECONDS, THE FIRE POINT TEMPERATURE IS REACHED.











Specific Gravity Test:

1) Density property is of great use to classify a binder. The density of bitumen is greatly influenced by its chemical composition.
2) The specific gravity of bitumen is defined as the ratio of mass of given volume of bitumen of known content to the mass of equal volume of water at 27^0C.
3) The specific gravity can be measured using either pycnometer or preparing a cube specimen of bitumen is semi solid or solid state.
4) The specific gravity of bitumen varies from 0.97 to 1.02.



Viscosity Test:

1) The fluid property of bituminous material & it is a measure of **resistance to flow**.
2) Low or **high viscosity** during compaction or mixing has been observed to result in **lower stability values**.
3) At **high viscosity,** it **resist the compactive effort** & thereby resulting mix is **heterogeneous**, hence **low stability values.**
4) **Orifice type viscometer** are used to indirectly find the viscosity of liquid binders like **cutbacks & emulsions.**
5) The viscosity expressed in seconds is the **time taken by the 50ml bitumen** material to pass through the orifice of a cup, under standard test conditions & specified temperature.
6) Viscosity of a cutback can be measured with either **4 mm orifice at 25^0C** or 10 mm orifice at 25^0C or 40^0C.





Test Name	Use / Purpose	Formula (if applicable)
Penetration Test	Measures the hardness or softness of bitumen by determining the depth (in 0.1 mm) a needle penetrates under specified conditions.	Penetration (d) = Depth of needle penetration (in 0.1 mm)
Ductility Test	Measures the distance in cm that a standard bitumen briquette can stretch before breaking under specified conditions.	Ductility (cm) = Final elongation length before break
Softening Point Test (Ring and Ball Method)	Determines the temperature at which bitumen softens under specific conditions.	Softening Point (°C) = Temperature at which steel ball touches base
Flash and Fire Point Test	Determines the temperature at which bitumen vapors momentarily ignite (Flash Point) and sustain burning (Fire Point).	No formula; Observed temperature readings
Specific Gravity Test	Measures the ratio of the density of bitumen to the density of water.	Specific Gravity (G) = Weight of bitumen / Weight of equal volume of water
Viscosity Test	Measures the resistance of bitumen to flow; important for workability and performance.	Absolute Viscosity (poises) = $(C \times t) / v$ Where: C = calibration constant t = time (seconds) v = volume (cm^3)

Bituminous mix design

The bituminous mix design aims to determine the proportion of bitumen, filler, fine aggregates, and coarse aggregates to produce a mix which is workable, strong, durable and economical

Objectives of mix design

The objective of the mix design is to produce a bituminous mix by proportionating various components so as
to have:

1. sufficient bitumen to ensure a durable pavement
2. sufficient strength to resist shear deformation under traffic at higher temperature
3. sufficient air voids in the compacted bitumen to allow for additional compaction by traffic
4. sufficient workability to permit easy placement without segregation,
5. sufficient flexibility to avoid premature cracking due to repeated bending by traffic
6. sufficient flexibility at low temperature to prevent shrinkage cracks.

Constituents of a Bituminous Mix

Constituent	Function	Examples
Coarse Aggregates	Provide compressive and shear strength; ensure good interlocking.	Granite
Fine Aggregates	Fill voids in coarse aggregates; stiffen the binder.	Sand, Rock dust
Filler	Fill voids, stiffen the binder, and improve permeability.	Rock dust, Cement, Lime
Binder	Cause particle adhesion, gluing, and provide impermeability.	Bitumen, Asphalt, Tar

Types of Bituminous Mix

Type of Mix	Characteristics
Well-graded Mix	Good proportion of all constituents; called Dense Bituminous Macadam; offers good compressive & tensile strength.
Gap-graded Mix	Some large aggregates missing; good fatigue and tensile strength.
Open-graded Mix	Fine aggregates and filler missing; porous; good friction; low strength; suitable for high-speed roads.
Unbounded	Binder absent; relies only on interlocking; very low tensile strength; requires kerb protection.

Different layers in a pavement

Layer	Material	Purpose	Key Characteristics
Bituminous Base Course	Mineral aggregate (stone, gravel, sand) + bituminous binder	Forms a strong foundation to support binder and surface courses.	Bonded by bitumen; provides stability and strength.
Bituminous Binder Course	Bituminous-aggregate mixture	Acts as an intermediate layer between base and surface courses; also used as the first layer in two-layer resurfacing.	Also called leveling course; improves surface evenness.
Asphaltic/ Bituminous Concrete	Continuously graded aggregates (<25 mm) + fine filler (<0.075 mm) + sufficient bitumen	Used as surface layer providing strength, durability, and smoothness.	Dense, *effectively impervious*; dissipative and elastic properties.

Table 24:1: Aggregate gradation for BC

Sieve size		Passing (%)
26.5	mm	- 100
19.	mm	90 - 100
9.5	mm	56 - 80
4.75	mm	35 - 65
2.36	mm	23 - 49
300	micron	5 - 19
75	micron	2 - 8

Requirements of Bituminous mixes

Property	Definition	Cause/Factors	Failure Examples
Stability	Resistance of mix to deformation under traffic load.	Inter-particle friction (aggregates) and cohesion (bitumen). Decreases with excessive binder.	Shoving (transverse deformation), Grooving (longitudinal ridging)
Durability	Resistance to weathering and abrasive actions.	Loss of volatiles (weathering), wheel load abrasion. Improved by higher binder content.	Pot-holes, Stripping (loss of binder)
Flexibility	Ability to bend and resist cracking due to traffic loads.	Binder shrinkage (aging), brittleness from repeated loads. Improved by higher binder content.	Shrinkage cracks, Alligator cracks
Skid Resistance	Ability to resist skidding by vehicles.	Surface texture (open graded) and bitumen content.	Skidding accidents (no cracks but poor grip)
Workability	Ease of laying, compacting, and shaping the mix.	Aggregate gradation, shape, texture, bitumen content/type. Rounded aggregates improve workability.	Poor compaction or uneven surface

Desirable properties

From the above discussion, the desirable properties of a bituminous mix can be summarized as follows:

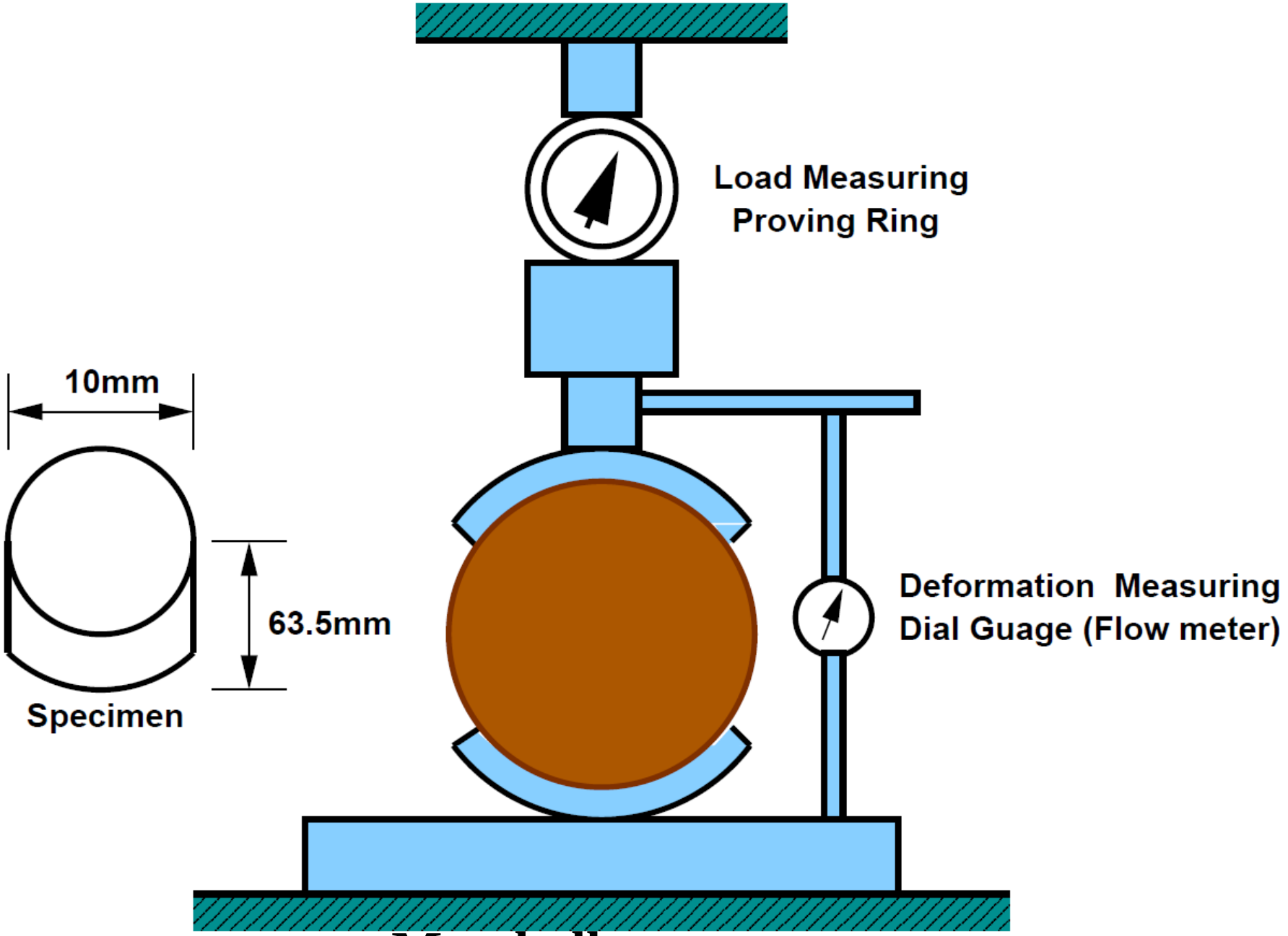
- Stability to meet traffic demand
- Bitumen content to ensure proper binding and water proofing
- Voids to accommodate compaction due to traffic
- Flexibility to meet traffic loads, esp. in cold season
- Sufficient workability for construction
- Economical mix

Marshall Mix Design

- There are many methods available for mix design which vary in the size of the test specimen, compaction, and other test specifications.
- Marshall method of mix design is the most popular one.
- The Marshall stability and flow test provides the performance prediction measure for the Marshall mix design method.
- The stability portion of the test measures the maximum load supported by the test specimen at a loading rate of 50.8 mm/minute.
- Load is applied to the specimen till failure, and the maximum load is designated as stability.
- During the loading, an attached dial gauge measures the specimen's plastic flow (deformation) due to the loading.
- The flow value is recorded in 0.25 mm (0.01 inch) increments at the same time when the maximum load is recorded.

Specimen preparation

1. Approximately 1200gm of aggregates and filler is heated to a temperature of 175 − 190^oC.
2. Bitumen is heated to a temperature of 121 − 125 C with the first trial percentage of bitumen (say 3.5 or 4% by weight of the mineral aggregates).
3. The heated aggregates and bitumen are thoroughly mixed at a temperature of 154 − 160 C. The mix is placed in a preheated mould and compacted by a rammer with 50 blows on either side at temperature of 138^oC to 149^oC.
4. The weight of mixed aggregates taken for the preparation of the specimen may be suitably altered to obtain a compacted thickness of 63.5+/-3 mm.
5. Vary the bitumen content in the next trial by +0.5% and repeat the above procedure. Number of trials are predetermined.
6. The prepared mould is loaded in the Marshall test setup as shown in the figure.



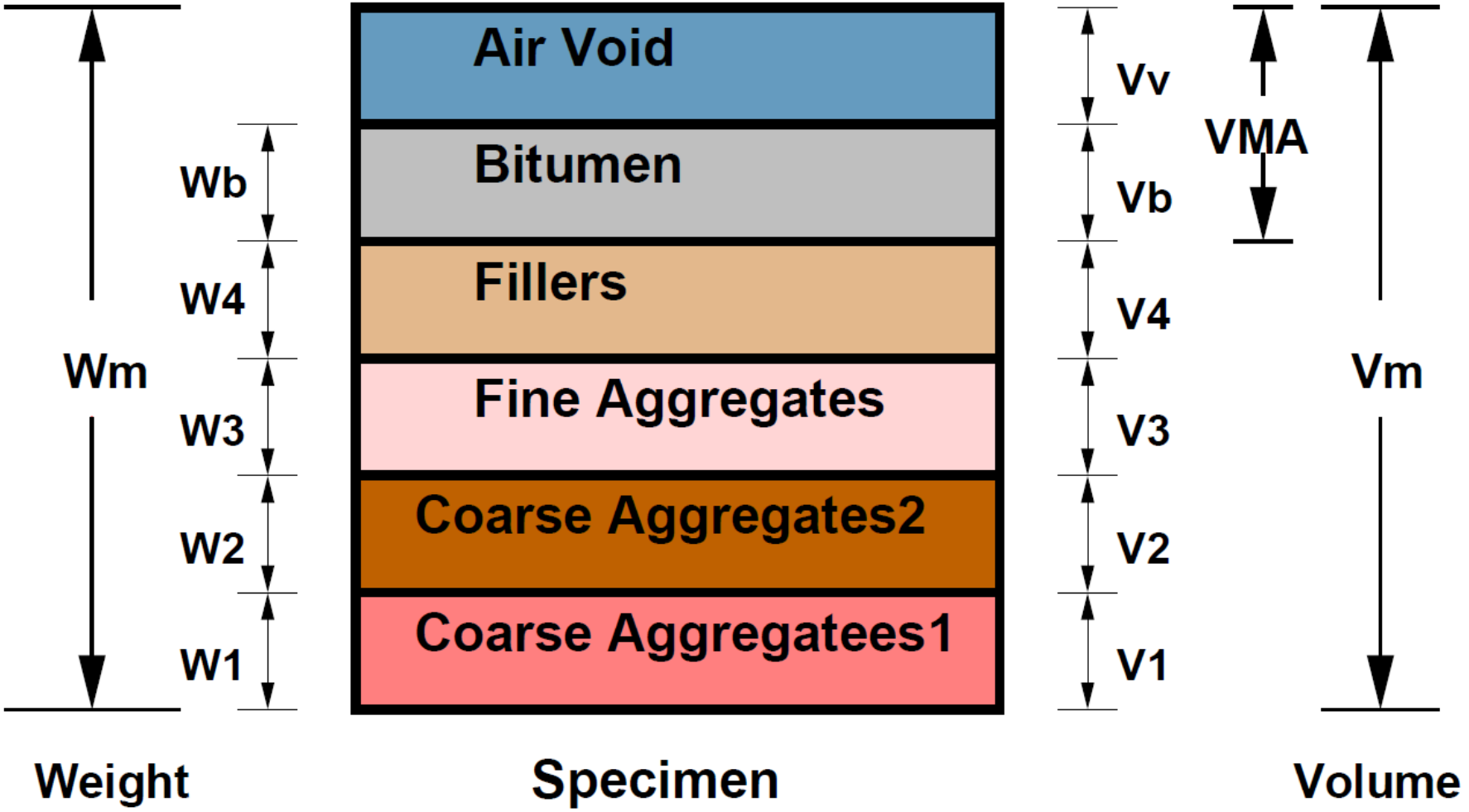
Marshall test setup

Properties of the mix

The properties that are of interest include

the theoretical specific gravity G_t, the bulk specific gravity of the mix G_m, percent air voids V_v, percent volume of bitumen V_b, percent void in mixed aggregate VMA , percent voids filled with bitumen VFB.

These calculations are discussed next. To understand these calculation a phase diagram is given in Figure



1) Theoretical specific gravity of the mix G_t

$$G_t = \frac{W_1 + W_2 + W_3 + W_b}{\frac{W_1}{G_1} + \frac{W_2}{G_2} + \frac{W_3}{G_3} + \frac{W_b}{G_b}}$$

where,

W_1 is the weight of coarse aggregate in the total mix,

W_2 is the weight of fine aggregate in the total mix,

W_3 is the weight of filler in the total mix,

W_b is the weight of bitumen in the total mix,

G_1 is the apparent specific gravity of coarse aggregate,

G_2 is the apparent specific gravity of fine aggregate,

G_3 is the apparent specific gravity of filler and

G_b is the apparent specific gravity of bitumen

2) Bulk specific gravity of mix Gm

The bulk specific gravity or the actual specific gravity of the mix G_m is the specific gravit considering air voids and is found out by

$$G_m = \frac{W_m}{W_m - W_w}$$

where,

W_m is the weight of mix in air,

W_w is the weight of mix in water,

Note that $W_m - W_w$ gives the volume of the mix. Sometimes to get accurate bulk specific gravity, the specimen is coated with thin film of paraffin wax, when weight is taken in the water.

This, however requires to consider the weight and volume of wax in the calculations.

3) Air voids percent V_v

Air voids V_v is the percent of air voids by volume in the specimen and is given by

$$V_v = \frac{(G_t - G_m)100}{G_t}$$

where G_t is the theoretical specific gravity of the mix,
G_m is the bulk or actual specific gravity of the mix

4) Percent volume of bitumen Vb

The volume of bitumen Vb is the percent of volume of bitumen to the total volume

$$V_b = \frac{\frac{W_b}{G_b}}{\frac{W_1+W_2+W_3+W_b}{G_m}}$$

where,
W1 is the weight of coarse aggregate in the total mix,
W2 is the weight of ne aggregate in the total mix,
W3 is the weight of filler in the total mix,
Wb is the weight of bitumen in the total mix,
Gb is the apparent specific gravity of bitumen,
Gm is the bulk specific gravity of mix

5) Voids in mineral aggregate VMA

Voids in mineral aggregate VMA is the volume of voids in the aggregates, and is the sum of air voids and volume of bitumen, and is calculated from

$$VMA = Vv + Vb$$

where,
Vv is the percent air voids in the mix
Vb is percent bitumen content in the mix

6)Voids filled with bitumen VFB

Voids filled with bitumen VFB is the voids in the mineral aggregate frame work filled with the bitumen, and is calculated as:

$$VFB = \frac{V_b \times 100}{VMA}$$

where,
Vb is percent bitumen content in the mix,.
VMA is the percent voids in the mineral aggregate

Determine Marshall stability and flow

- Marshall stability of a test specimen is the maximum load required to produce failure when the specimen is preheated to a prescribed temperature placed in a special test head and the load is applied at a constant strain (5 cm per minute).
- While the stability test is in progress dial gauge is used to measure the vertical deformation of the specimen.
- The deformation at the failure point expressed in units of 0.25 mm is called the Marshall flow value of the specimen.

Table 26:2: Marshall mix design specification

Test Property	Specified Value
Marshall stability, kg	340 (minimum)
Flow value, 0.25 mm units	8 - 17
Percent air voids in the mix $V_v\%$	3 - 5
Voids filled with bitumen $VFB\%$	75 - 85

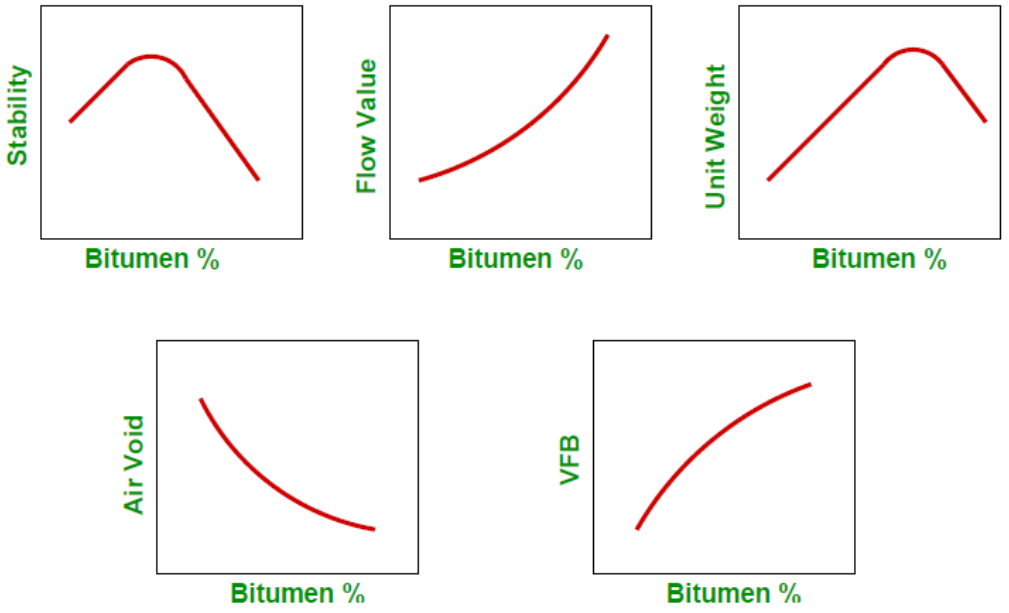


Figure 26:3: Marshal graphical plots

Numerical example

The specific gravities and weight proportions for aggregate and bitumen are as under for the preparation of Marshall mix design. The volume and weight of one Marshall specimen was found to be 475 cc and 1100 gm. Assuming absorption of bitumen in aggregate is zero, find V_v, V_b, *VMA* and *VFB*.

Item	A_1	A_2	A_3	A_4	B
Wt (gm)	825	1200	325	150	100
Sp. Gr	2.63	2.51	2.46	2.43	1.05

$$
\begin{aligned}
G_t &= \frac{825 + 1200 + 325 + 150 + 100}{\frac{825}{2.63} + \frac{1200}{2.51} + \frac{325}{2.46} + \frac{150}{2.43} + \frac{100}{1.05}} \\
&= \frac{2600}{1080.86} \\
&= 2.406 \\
G_m &= \frac{1100}{475} \\
&= 2.316 \\
V_v &= \frac{2.406 - 2.316}{2.406} \times 100 \\
&= 3.741\ \% \\
V_b &= \frac{100}{1.05} \times \frac{2.316}{1100} \\
&= 20.052\ \% \\
VMA &= (3.741 + 20.05) \\
&= 23.793\ \% \\
VFB &= \frac{20.052}{23.793} \times 100 \\
&= 84.277\ \%
\end{aligned}
$$